

Soft Spaces in Low-Qubit Hilbert Spaces: A 4-to-8 Qubit Study of REAL–NULL Separation, Subspace Stability, and Dynamical Structure

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Abstract

This paper presents a numerical investigation of structured subspaces in low-qubit Hilbert spaces. The study is motivated by the hypothesis that computationally or dynamically relevant structure may not be distributed uniformly across the full Hilbert space, but may instead concentrate in selected regions. Such regions are here referred to operationally as *soft spaces*: candidate subspaces whose behaviour differs reproducibly from matched control systems.

The investigation is based on a REAL-versus-NULL methodology. For each qubit number n , observed structures in REAL configurations are compared with spectrum-matched NULL controls. The central evidential quantity is the contrast

$$\Delta M(n) = M_{\text{REAL}}(n) - M_{\text{NULL}}(n),$$

where M denotes a static or dynamical metric. Static metrics include separation scores, entropy-related structure, ranked family signatures, and batch recurrence. Dynamical metrics include retention, leakage, local purity, and structure drift under open-system evolution.

The study follows a controlled scaling track from 4 to 8 qubits. This corresponds to Hilbert-space dimensions increasing from

$$2^4 = 16$$

to

$$2^8 = 256.$$

The purpose of this scaling track is to determine whether candidate soft-space structures are isolated low-dimensional artefacts, or whether they persist and transform as the Hilbert space expands. Particular attention is given to whether lower-qubit structures can act as priors for higher-qubit searches, leading to the identification of recurrent anchor regions and candidate hotspots.

The main empirical result is that static REAL – NULL separation is stronger and more stable than the corresponding dynamical signal. Across the 4-to-8 qubit sequence, selected static metrics show reproducible structure and batch-level recurrence. Dynamical tests based on Lindblad-type open-system evolution show weaker effects, but in several cases the direction of the signal remains consistent: REAL-favoured regions tend to show slightly improved retention, reduced leakage, higher local purity, or lower structural drift compared with matched controls.

These findings motivate a shift from uniform scanning of Hilbert space to targeted subspace-selective analysis. The results suggest that static separation can be used as a prior for identifying dynamically promising regions, while Lindblad-based evolution provides a second filter for testing whether such regions retain structure under noise. The paper

therefore frames the 4-to-8 qubit study as a Phase 1 evidence base for later hierarchical searches, including v24_8-style dynamic-kernel identification and subsequent anchor-based refinement.

Keywords: quantum information; Hilbert space; low-qubit systems; open quantum systems; Lindblad dynamics; REAL–NULL controls; subspace selection; soft spaces; NISQ; quantum noise; dynamical kernels; anchor points.

1 Introduction

Low-qubit quantum systems are often treated as too small to contain practically meaningful internal structure beyond the directly specified computational basis. However, even modest Hilbert spaces grow exponentially with the number of qubits. A system of n qubits spans a complex Hilbert space of dimension

$$\dim(\mathcal{H}_n) = 2^n.$$

For the qubit range considered in this paper,

$$n \in \{4, 5, 6, 7, 8\},$$

the corresponding Hilbert-space dimensions are

$$2^n \in \{16, 32, 64, 128, 256\}.$$

Although these dimensions are small compared with large-scale fault-tolerant quantum architectures, they are already large enough to support non-trivial internal subspace structure. The central motivation of this work is therefore to examine whether such low-qubit Hilbert spaces should be treated as uniform computational arenas, or whether selected regions inside them exhibit measurably different behaviour.

The project is based on the hypothesis that relevant structure may not be evenly distributed across the full Hilbert space. Instead, structure may concentrate in particular subspaces. These regions may show stronger statistical separation, greater recurrence across batches, improved retention under open-system evolution, reduced leakage, higher local purity, or lower structural drift. The purpose of the present study is to test whether such regions can be detected systematically by comparing observed configurations with matched controls.

In this paper, such candidate regions are referred to operationally as *soft spaces*. A soft space is not assumed to be a new physical object or a new phase of matter. Rather, the term denotes an empirically identified subregion of Hilbert space whose measured behaviour differs reproducibly from an appropriate null model. In the present framework, a soft-space candidate becomes credible only if it shows consistent REAL – NULL separation under pre-specified metrics.

The relevant contrast is written generically as

$$\Delta M(n) = M_{\text{REAL}}(n) - M_{\text{NULL}}(n),$$

where M may denote a static metric, such as entropy-related structure or family recurrence, or a dynamical metric, such as retention, leakage, purity, or drift under open-system evolution. The NULL systems are designed to preserve selected baseline properties while removing or randomising the structure under investigation. This makes the REAL – NULL difference the primary evidential quantity throughout the study.

A central distinction in the paper is the distinction between static and dynamical evidence. Static evidence concerns structure measured before open-system evolution, for example ranked separation, entropy shifts, or recurrent signature families. Dynamical evidence concerns how

selected regions behave under noise or dissipation, modelled here through Lindblad-type open-system dynamics. The strongest evidence obtained so far lies in static REAL – NULL separation. The dynamical signal is weaker, but in selected cases it is directionally consistent, suggesting that some statistically favoured regions may also possess limited dynamical robustness.

This paper reports a staged numerical investigation from 4 to 8 qubits. The aim is not to prove universal scalability, nor to claim the construction of fault-tolerant logical qubits. The aim is more limited and methodological: to document whether a stable evidence trail exists across increasing Hilbert-space dimension, and whether lower-qubit structures can be used as priors for higher-qubit searches.

The study therefore has the character of a Phase 1 investigation. It establishes the basic REAL – NULL methodology, evaluates the persistence of static separation from 4 to 8 qubits, examines weaker dynamical indicators under open-system evolution, and motivates a transition from uniform Hilbert-space scanning to targeted subspace-selective analysis. In later versions of the project, including v24_8 and beyond, this transition becomes central: the objective is no longer merely to detect whether structure exists, but to identify where dynamically promising kernels are concentrated.

2 Research Question and Hypothesis

The central research question of this paper is:

Can low-qubit Hilbert spaces contain identifiable subspaces whose statistical, structural, or dynamical behaviour differs consistently from spectrum-matched NULL controls?

This question is deliberately formulated in operational rather than ontological terms. The aim is not to assume in advance that soft spaces exist as independent physical objects. Instead, the aim is to test whether selected subregions of Hilbert space can be identified by reproducible metric differences between REAL configurations and matched NULL controls.

For an n -qubit system, let

$$\mathcal{H}_n \cong \mathbb{C}^{2^n}$$

denote the Hilbert space, and let

$$\mathcal{S}_{n,j} \subset \mathcal{H}_n$$

denote the j -th candidate subspace at qubit number n . The working hypothesis is that for some n and j , the candidate subspace $\mathcal{S}_{n,j}$ exhibits measurable behaviour that differs from the corresponding NULL model.

A generic REAL-NULL contrast may be written as

$$\Delta M_{n,j} = M_{\text{REAL}}(n, j) - M_{\text{NULL}}(n, j),$$

where M is a pre-specified metric. Depending on the test, M may represent a static quantity, such as entropy-related structure, family recurrence, or ranked separation, or a dynamical quantity, such as retention, leakage, purity, or structure drift.

The soft-space hypothesis can therefore be stated as follows:

A candidate soft space exists if there is at least one subspace $\mathcal{S}_{n,j}$ for which $\Delta M_{n,j}$ is reproducibly non-zero, directionally stable across batches, and interpretable under a pre-specified metric family.

More specifically, a candidate subspace becomes stronger if it satisfies several of the following conditions:

1. it shows positive static REAL-NULL separation;
2. it recurs across independent batches or seeds;
3. it belongs to a recurrent signature family or anchor neighbourhood;
4. it retains more probability mass under open-system evolution;
5. it exhibits lower leakage from the selected subspace;
6. it maintains higher local purity;
7. it shows lower structural drift under Lindblad-type dynamics.

A relevance-like quantity may be written schematically as

$$R_{n,j} = R(\mathcal{S}_{n,j}),$$

where R denotes the measured relevance of candidate subspace $\mathcal{S}_{n,j}$. In the simplest inside–outside formulation, one may define

$$\Delta R(n, j) = R_{\text{inside}}(n, j) - R_{\text{outside}}(n, j),$$

where $R_{\text{inside}}(n, j)$ denotes the measured relevance inside the candidate subspace. The quantity $R_{\text{outside}}(n, j)$ denotes the corresponding relevance outside it.

In the REAL-NULL formulation, the more important contrast is

$$\Delta R_{\text{REAL-NULL}}(n, j) = R_{\text{REAL}}(n, j) - R_{\text{NULL}}(n, j).$$

The preferred case is therefore a candidate subspace for which

$$\Delta R_{\text{REAL-NULL}}(n, j) > 0,$$

provided that the sign, magnitude, and recurrence of the effect remain stable under repeated runs.

The exact operational definition of R remains a developing component of the framework. In this paper, R should therefore be understood not as a single final scalar, but as a controlled metric family. Candidate definitions include fidelity-based retention, probability-density concentration, reduced leakage, higher local purity, lower dynamical drift, computational cost reduction, or composite scores combining several of these quantities.

This distinction is important. If R were left undefined, the concept of relevance would remain too vague to support a reproducible test. By treating R as a metric family, the framework allows different operational definitions to be compared explicitly. A candidate soft space is therefore not accepted merely because it looks visually or numerically interesting. It must survive comparison with matched NULL controls under one or more clearly specified metrics.

The hypothesis tested in this paper is consequently modest but concrete: low-qubit Hilbert spaces may contain subspaces whose measured behaviour is not fully explained by the matched NULL construction. The strongest expected evidence is reproducible static REAL-NULL separation. A stronger and more selective form of evidence occurs when the same or related subspaces also show favourable behaviour under Lindblad-type dynamical evolution.

3 Conceptual Background

3.1 Hilbert-space scaling

An n -qubit pure state can be written as

$$|\psi\rangle = \sum_{i=0}^{2^n-1} \alpha_i |i\rangle,$$

where $|i\rangle$ denotes a computational-basis state and the complex amplitudes α_i satisfy the normalisation condition

$$\sum_{i=0}^{2^n-1} |\alpha_i|^2 = 1.$$

The associated Hilbert space is

$$\mathcal{H}_n \cong \mathbb{C}^{2^n}.$$

The exponential dependence on n means that each additional qubit doubles the dimension of the available state space. In the present study, the qubit sequence

$$n \in \{4, 5, 6, 7, 8\}$$

corresponds to the dimensional progression

$$16 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256.$$

This progression is important because the move from 4 to 8 qubits is not merely an addition of four physical two-level systems. It is a sixteen-fold increase in Hilbert-space dimension. As the dimension grows, the number of possible subspaces, neighbourhoods, signature families, and candidate anchor structures grows rapidly. Consequently, the search for structured regions becomes increasingly non-trivial.

The central methodological implication is that a uniform scan of the full Hilbert space becomes less informative as n increases. If relevant structure is concentrated in selected regions, then the main problem is not only to measure global behaviour, but to identify which internal regions carry the strongest evidence. This motivates a transition from global Hilbert-space comparison to subspace-selective analysis.

3.2 REAL and NULL systems

The project compares observed or constructed REAL configurations against matched NULL controls. The purpose of the NULL model is to preserve selected baseline properties while removing or randomising the specific structure under investigation. In this way, the NULL system provides a reference point for deciding whether an observed pattern is meaningful or merely an artefact of dimension, spectrum, sampling, or noise.

For a given metric M , the basic evidential contrast is

$$\Delta M = M_{\text{REAL}} - M_{\text{NULL}}.$$

When the metric depends on qubit number n and candidate subspace j , this may be written more explicitly as

$$\Delta M_{n,j} = M_{\text{REAL}}(n, j) - M_{\text{NULL}}(n, j).$$

Here, M may denote entropy-related structure, ranked separation, family recurrence, fidelity retention, leakage, local purity, or structural drift. The interpretation of the sign of $\Delta M_{n,j}$ depends on the metric. For retention, purity, or relevance, a positive value is typically favourable. For leakage or drift, a negative value is typically favourable.

The use of matched NULL controls is essential because a raw signal inside a Hilbert space is not sufficient evidence of structure. A high score may arise simply because of the dimension of the system, the sampling procedure, the spectral profile, or the noise model. A candidate soft space becomes meaningful only if the corresponding REAL region separates from an appropriate NULL control under pre-specified metrics.

Thus, the main evidential object in the paper is not an isolated metric value, but the reproducible contrast between REAL and NULL:

$$\text{evidence} \iff \Delta M_{n,j} \text{ is stable, interpretable, and recurrent.}$$

3.3 Static versus dynamical evidence

The project distinguishes between static and dynamical evidence.

Static evidence concerns structure measured before open-system evolution. This includes quantities such as entropy shifts, ranked score separation, signature-family recurrence, overlap structure, and batch-level stability. Static evidence answers the question: does a selected region of Hilbert space already differ from its matched NULL counterpart before noise or dissipation is applied?

Dynamical evidence concerns how selected states or subspaces behave under open-system evolution. In this paper, such evolution is modelled through Lindblad-type dynamics. Dynamical evidence answers a stronger question: does a candidate region retain favourable structure when exposed to noise, leakage, decoherence, or dissipative evolution?

The distinction can be represented schematically as

$$\Delta M_{\text{static}}(n, j) = M_{\text{static,REAL}}(n, j) - M_{\text{static,NULL}}(n, j),$$

and

$$\Delta M_{\text{dyn}}(n, j, t) = M_{\text{dyn,REAL}}(n, j, t) - M_{\text{dyn,NULL}}(n, j, t).$$

The first expression compares static structure at baseline. The second compares time-dependent behaviour under open-system evolution. A time-aggregated dynamical contrast may also be defined as

$$\overline{\Delta M}_{\text{dyn}}(n, j) = \frac{1}{T} \int_0^T \Delta M_{\text{dyn}}(n, j, t) dt.$$

In numerical experiments, this integral is approximated by a discrete time average or an area-under-curve estimate.

The current evidence trail suggests an asymmetry between these two types of evidence. Static REAL – NULL separation has been stronger and more reproducible. Dynamical separation has been weaker in magnitude, but in selected cases it remains directionally consistent. This asymmetry is important for the interpretation of the project. The static results provide the primary evidence for candidate soft spaces, while the dynamical results act as a second and more selective filter.

This leads to the working interpretation used throughout the paper: static separation identifies candidate regions, whereas Lindblad-type dynamical testing evaluates whether those regions have any degree of robustness under open-system evolution. A candidate region that satisfies both criteria is substantially more interesting than one that separates only statically.

4 Lindblad Dynamics as a Scaling Framework from 4 to 8 Qubits

The purpose of this section is to develop the Lindblad-based part of the framework as a scaling tool from 4 to 8 qubits. The aim is not only to describe open-system evolution at a fixed qubit number, but to evaluate whether candidate soft-space structures can be followed as the Hilbert-space dimension increases.

The section proceeds in six steps. First, the Lindblad equation is written explicitly for an n -qubit system. Second, the transition from n to $(n+1)$ qubits is formulated as an embedding and continuation problem. Third, the dynamical metrics used to evaluate candidate regions are defined. Fourth, anchor points and hotspots are treated as possible recurrent structural coordinates. Fifth, the 4-to-8 qubit progression is interpreted stepwise. Finally, these elements are combined into a preliminary predictive scaling rule.

4.1 Lindblad evolution for an n -qubit system

For an n -qubit system, the state space is the Hilbert space

$$\mathcal{H}_n \cong \mathbb{C}^{2^n}.$$

A pure state is represented by a vector $|\psi_n\rangle \in \mathcal{H}_n$, while a general mixed or open-system state is represented by a density matrix

$$\rho_n(t) \in \mathbb{C}^{2^n \times 2^n}.$$

The density matrix satisfies

$$\rho_n(t) \succeq 0,$$

and

$$\text{Tr}(\rho_n(t)) = 1.$$

Here, the subscript n is important. When the system is enlarged from 4 to 5, 6, 7, and 8 qubits, the dimension of $\rho_n(t)$ changes as

$$16 \times 16 \rightarrow 32 \times 32 \rightarrow 64 \times 64 \rightarrow 128 \times 128 \rightarrow 256 \times 256.$$

Thus, the move from 4 to 8 qubits does not merely add four extra physical degrees of freedom. It changes the size of the dynamical object from a 16×16 density matrix to a 256×256 density matrix.

The open-system evolution is modelled by a Lindblad master equation,

$$\frac{d\rho_n(t)}{dt} = -i[H_n, \rho_n(t)] + \sum_k \left(L_{k,n} \rho_n(t) L_{k,n}^\dagger - \frac{1}{2} \{ L_{k,n}^\dagger L_{k,n}, \rho_n(t) \} \right).$$

In this expression, H_n is the effective Hamiltonian of the n -qubit system, and the operators $L_{k,n}$ describe the noise or dissipative channels acting on the system. The commutator term

$$-i[H_n, \rho_n(t)]$$

describes the coherent part of the evolution, while the remaining terms describe the irreversible open-system contribution.

The Lindblad operators $L_{k,n}$ depend on the chosen noise model. For example, a dephasing-type channel may be represented by operators proportional to Pauli- Z operators acting on selected qubits, while an amplitude-damping-type channel may be represented by lowering operators acting on selected qubits. In the present framework, the specific choice of noise model is not

treated as a secondary detail. It determines which aspects of a candidate soft space are being stress-tested.

The n -dependence of the Lindblad equation is essential for the 4-to-8 qubit programme. At each qubit number, the Hamiltonian, the density matrix, the noise operators, and the candidate subspaces all live in a different-dimensional Hilbert space:

$$\rho_4(t) \in \mathbb{C}^{16 \times 16},$$

$$\rho_5(t) \in \mathbb{C}^{32 \times 32},$$

$$\rho_6(t) \in \mathbb{C}^{64 \times 64},$$

$$\rho_7(t) \in \mathbb{C}^{128 \times 128},$$

and

$$\rho_8(t) \in \mathbb{C}^{256 \times 256}.$$

Consequently, a dynamical pattern observed at 4 qubits cannot simply be assumed to persist at 5, 6, 7, or 8 qubits. It must be re-evaluated after each dimensional enlargement.

The role of Lindblad dynamics in this paper is therefore twofold. First, it provides a physically motivated model for testing how candidate subspaces behave under noise, dissipation, and decoherence. Second, it provides a scaling framework for asking whether structures found at lower qubit numbers have dynamically meaningful continuations at higher qubit numbers.

In this interpretation, a candidate soft space is not validated merely by having a strong static score. It must also be tested dynamically. A region that separates strongly from its matched NULL control at baseline may still lose its structure rapidly under open-system evolution. Conversely, a region with only moderate static separation may become more interesting if it shows unusually high retention, low leakage, higher local purity, or reduced drift under Lindblad evolution.

Thus, the Lindblad equation functions as a second evidential filter. Static analysis identifies candidate regions. Lindblad evolution tests whether those regions retain meaningful structure over time. The central scaling question is then whether the same type of dynamically favourable behaviour can be followed through the sequence

$$4 \rightarrow 5 \rightarrow 6 \rightarrow 7 \rightarrow 8$$

qubits.

4.2 Embedding from n to $(n+1)$ qubits

To compare structures across different qubit numbers, one must specify how an n -qubit Hilbert space is related to an $(n+1)$ -qubit Hilbert space. Since

$$\mathcal{H}_n \cong \mathbb{C}^{2^n}$$

and

$$\mathcal{H}_{n+1} \cong \mathbb{C}^{2^{n+1}},$$

the transition from n to $(n+1)$ qubits doubles the Hilbert-space dimension:

$$\dim(\mathcal{H}_{n+1}) = 2 \dim(\mathcal{H}_n).$$

A simple way to represent this enlargement is by tensoring the original space with an additional two-dimensional qubit space:

$$\mathcal{H}_{n+1} \cong \mathbb{C}^2 \otimes \mathcal{H}_n.$$

Thus, an n -qubit basis state $|x\rangle_n$ may be embedded into the $(n+1)$ -qubit space as

$$|x\rangle_n \mapsto |0\rangle_1 \otimes |x\rangle_n,$$

or, equivalently,

$$|x\rangle_n \mapsto |0x\rangle_{n+1}.$$

A second possible embedding is

$$|x\rangle_n \mapsto |1\rangle_1 \otimes |x\rangle_n = |1x\rangle_{n+1}.$$

These two embeddings correspond to placing the original n -qubit structure inside one of the two computational halves of the $(n+1)$ -qubit Hilbert space.

For a general state

$$|\psi_n\rangle = \sum_{i=0}^{2^n-1} \alpha_i |i\rangle_n,$$

the zero-branch embedding is

$$E_n^{(0)} |\psi_n\rangle = |0\rangle \otimes |\psi_n\rangle = \sum_{i=0}^{2^n-1} \alpha_i |0i\rangle_{n+1},$$

while the one-branch embedding is

$$E_n^{(1)} |\psi_n\rangle = |1\rangle \otimes |\psi_n\rangle = \sum_{i=0}^{2^n-1} \alpha_i |1i\rangle_{n+1}.$$

At the density-matrix level, an n -qubit state may be embedded as

$$\rho_n \mapsto \rho_{n+1}^{(0)} = |0\rangle \langle 0| \otimes \rho_n,$$

or as

$$\rho_n \mapsto \rho_{n+1}^{(1)} = |1\rangle \langle 1| \otimes \rho_n.$$

These formulae provide a minimal mathematical representation of how a structure found at n qubits can be placed inside the larger $(n+1)$ -qubit space.

However, the central object in the present project is not only a state, but a candidate subspace. If

$$\mathcal{S}_{n,j} \subset \mathcal{H}_n$$

is a candidate soft-space region at n qubits, then its direct embedded descendants may be written as

$$E_n^{(0)} (\mathcal{S}_{n,j}) \subset \mathcal{H}_{n+1},$$

and

$$E_n^{(1)} (\mathcal{S}_{n,j}) \subset \mathcal{H}_{n+1}.$$

These are literal embeddings. They preserve the internal structure of the original subspace but place it inside a larger Hilbert space. In practice, however, a soft-space continuation need not be a perfectly literal embedding. The relevant $(n+1)$ -qubit structure may be shifted, split, broadened, or concentrated relative to the embedded n -qubit structure.

For this reason, the project treats continuation from n to $(n+1)$ qubits as a structural matching problem. We write

$$\mathcal{S}_{n,j} \longrightarrow \hat{\mathcal{S}}_{n+1,j'}$$

to denote a predicted or observed continuation of the candidate subspace $\mathcal{S}_{n,j}$ at the next qubit number.

The continuation is not defined solely by basis-index inheritance. Instead, it may depend on a combination of static and dynamical evidence:

$$\hat{\mathcal{S}}_{n+1,j'} = \text{Cont}(\mathcal{S}_{n,j}; \Delta M_{\text{static}}, \Delta R, \Delta \Lambda, \Delta P, \Delta D, B).$$

Here, Cont denotes a continuation rule, ΔM_{static} is the static REAL – NULL contrast, ΔR is a retention contrast, $\Delta \Lambda$ is a leakage contrast, ΔP is a purity contrast, ΔD is a drift contrast, and B is a batch-recurrence or stability score.

The purpose of the embedding formalism is therefore not to claim that a soft-space structure must reappear at exactly doubled basis indices. Rather, the embedding gives a reference geometry. It defines where one would expect a direct descendant to appear if the lower-dimensional structure were simply carried into the larger space. Deviations from this direct embedding may themselves be informative, especially if the displaced or transformed structure shows stronger REAL – NULL separation.

For the 4-to-8 qubit programme, the relevant sequence is

$$\mathcal{H}_4 \rightarrow \mathcal{H}_5 \rightarrow \mathcal{H}_6 \rightarrow \mathcal{H}_7 \rightarrow \mathcal{H}_8,$$

with dimensions

$$16 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256.$$

At each step, candidate structures from the previous qubit number can be used as priors for the next search. The methodological question is then whether the structures discovered at $(n+1)$ qubits are compatible with the lower-qubit evidence, or whether they represent unrelated high-dimensional artefacts.

This gives a practical interpretation of the scaling programme:

$$\text{lower-qubit structure} \implies \text{higher-qubit search prior}.$$

In this sense, the transition from n to $(n+1)$ qubits is not merely a larger numerical experiment. It is an attempt to determine whether candidate soft-space structures have continuations, descendants, or refinements as the Hilbert-space dimension increases.

4.3 Retention, leakage, purity, and drift

Once a candidate subspace has been identified at a given qubit number, the next question is whether that subspace retains meaningful structure under open-system evolution. In the present framework, four dynamical quantities are especially important: retention, leakage, local purity, and structure drift.

Let

$$\mathcal{S}_{n,j} \subset \mathcal{H}_n$$

denote the j -th candidate subspace at qubit number n , and let

$$P_{n,j}$$

denote the orthogonal projector onto $\mathcal{S}_{n,j}$. The evolved density matrix is

$$\rho_n(t),$$

where $\rho_n(t)$ follows the Lindblad equation introduced above.

The retention of the state inside the candidate subspace is defined as

$$R_{n,j}(t) = \text{Tr}(P_{n,j}\rho_n(t)).$$

This quantity measures the amount of probability mass that remains inside the selected subspace at time t . If

$$R_{n,j}(t) \approx 1,$$

then most of the state remains inside the candidate region. If

$$R_{n,j}(t) \approx 0,$$

then the state has largely left the selected region.

The corresponding leakage is defined as

$$\Lambda_{n,j}(t) = 1 - R_{n,j}(t).$$

Thus, high retention and low leakage are equivalent descriptions of favourable dynamical confinement:

$$R_{n,j}(t) \uparrow \iff \Lambda_{n,j}(t) \downarrow.$$

For a REAL-versus-NULLE comparison, the retention contrast is

$$\Delta R_{n,j}(t) = R_{\text{REAL}}(n, j, t) - R_{\text{NULL}}(n, j, t),$$

and the leakage contrast is

$$\Delta \Lambda_{n,j}(t) = \Lambda_{\text{REAL}}(n, j, t) - \Lambda_{\text{NULL}}(n, j, t).$$

The preferred directions are

$$\Delta R_{n,j}(t) > 0,$$

and

$$\Delta \Lambda_{n,j}(t) < 0.$$

In words, a dynamically favourable REAL subspace should retain more probability mass and leak less than its matched NULL counterpart.

A second important dynamical quantity is purity. For the full density matrix, purity is

$$P_n(t) = \text{Tr}(\rho_n(t)^2).$$

A pure state has

$$P_n(t) = 1,$$

whereas a mixed state has lower purity. Under noise and dissipation, purity will typically decrease, although the precise behaviour depends on the Lindblad operators and the initial state.

For a candidate subspace $\mathcal{S}_{n,j}$, one may define the projected and renormalised density matrix

$$\rho_{n,j}^{\text{proj}}(t) = \frac{P_{n,j}\rho_n(t)P_{n,j}}{\text{Tr}(P_{n,j}\rho_n(t))},$$

provided that

$$\text{Tr}(P_{n,j}\rho_n(t)) \neq 0.$$

The local purity inside the candidate subspace is then

$$P_{n,j}(t) = \text{Tr} \left[\left(\rho_{n,j}^{\text{proj}}(t) \right)^2 \right].$$

The corresponding REAL – NULL purity contrast is

$$\Delta P_{n,j}(t) = P_{\text{REAL}}(n, j, t) - P_{\text{NULL}}(n, j, t).$$

The preferred direction is

$$\Delta P_{n,j}(t) > 0.$$

This means that a REAL-favoured subspace is more dynamically interesting if it maintains higher local purity than its matched NULL counterpart.

The fourth dynamical quantity is structure drift. Retention and purity measure whether probability mass and state concentration are preserved. Drift measures whether the internal structural descriptor of a candidate region changes over time.

Let

$$S_{n,j}(t)$$

denote a structural descriptor of the candidate subspace at time t . Depending on the implementation, this descriptor may represent a signature vector, a ranked family profile, an overlap pattern, a coherence profile, or another reduced representation of the local structure. A generic drift measure may be written as

$$D_{n,j}(t) = d(S_{n,j}(0), S_{n,j}(t)),$$

where $d(\cdot, \cdot)$ is a chosen distance function. Examples include Euclidean distance, cosine distance, total variation distance, or another implementation-specific discrepancy measure.

For a REAL – NULL comparison, the drift contrast is

$$\Delta D_{n,j}(t) = D_{\text{REAL}}(n, j, t) - D_{\text{NULL}}(n, j, t).$$

The preferred direction is

$$\Delta D_{n,j}(t) < 0.$$

A negative drift contrast means that the REAL structure changes less over time than the corresponding NULL structure.

Since Lindblad evolution is time-dependent, the project often requires time-aggregated versions of these quantities. For a generic dynamical metric $X_{n,j}(t)$, define the time average

$$\overline{X}_{n,j} = \frac{1}{T} \int_0^T X_{n,j}(t) dt.$$

In numerical experiments, this is approximated by a discrete average over sampled time points,

$$\bar{X}_{n,j} \approx \frac{1}{m} \sum_{\ell=1}^m X_{n,j}(t_\ell),$$

where

$$0 \leq t_1 < t_2 < \dots < t_m \leq T.$$

Similarly, one may define area-under-curve quantities such as

$$\text{AUC}(R_{n,j}) = \int_0^T R_{n,j}(t) dt,$$

with analogous definitions for leakage, purity, and drift.

The four dynamical metrics can now be summarised by their preferred REAL – NULL directions:

$$\Delta R_{n,j} > 0,$$

$$\Delta \Lambda_{n,j} < 0,$$

$$\Delta P_{n,j} > 0,$$

and

$$\Delta D_{n,j} < 0.$$

Taken together, these conditions define a candidate dynamically favourable soft-space region. Such a region should retain more probability mass, leak less, maintain higher local purity, and drift less than its matched NULL control.

In the current evidence trail, the dynamical indicators are weaker than the static REAL – NULL separation. This does not make them irrelevant. Rather, it changes their role. Static metrics identify candidate regions, while retention, leakage, purity, and drift act as a second filter. A subspace that is both statically separated and dynamically favourable is a stronger candidate than a subspace that is only statically unusual.

4.4 Anchor points and hotspot continuation

As the Hilbert-space dimension increases, candidate soft-space structures become more difficult to identify by global inspection alone. For this reason, the project introduces the idea of anchor points and hotspots. These are not assumed to be fundamental physical constants. They are operational markers: recurrent coordinates, index pairs, neighbourhoods, or signature locations where REAL-favoured structure appears more often or more strongly than expected under matched NULL controls.

Let

$$a_{n,j}$$

denote an anchor associated with candidate subspace $\mathcal{S}_{n,j}$ at qubit number n . In the simplest case, an anchor may be represented by a basis index

$$a_{n,j} = i,$$

or by an index pair

$$a_{n,j} = (i, k).$$

More generally, an anchor may represent a local neighbourhood

$$\mathcal{N}_n(a_{n,j}) \subset \mathcal{H}_n,$$

where $\mathcal{N}_n(a_{n,j})$ denotes the region around the anchor in index space, signature space, overlap space, or another implementation-specific representation.

The purpose of an anchor is to provide a reference point for tracking structure across batches, noise models, and qubit numbers. A hotspot is then a candidate anchor neighbourhood where the measured REAL – NULL contrast is especially strong. In this sense, a hotspot is not merely a high-scoring point. It is a high-scoring point or region that remains meaningful under repeated comparison with matched controls.

For a metric M , a hotspot condition may be written schematically as

$$\Delta M_{n,j} = M_{\text{REAL}}(n, j) - M_{\text{NULL}}(n, j)$$

with

$$|\Delta M_{n,j}|$$

large relative to the background distribution of candidate subspaces. The preferred sign depends on the metric. For retention, purity, or relevance, the preferred direction is positive. For leakage or drift, the preferred direction is negative.

A stronger condition is recurrence. Let

$$B_{n,j}$$

denote the batch-recurrence score of candidate anchor $a_{n,j}$. If there are B independent batches, one may define

$$B_{n,j} = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{a_{n,j} \text{ appears as a candidate in batch } b\}.$$

A candidate anchor is more credible when $B_{n,j}$ is high. In particular, a hotspot that appears in only one run may be an artefact of sampling, whereas a hotspot that recurs across batches is more likely to reflect stable structure.

In the 8-qubit stage of the project, recurrent candidates such as

$$(157, 158)$$

and nearby or related index pairs have appeared as possible anchor structures. Such pairs should not be interpreted as meaningful merely because of their numerical labels. Their significance depends on whether they satisfy several conditions:

1. they are associated with favourable REAL – NULL separation;
2. they recur across batches or repeated runs;
3. they belong to a coherent local neighbourhood rather than an isolated accidental index;
4. they remain compatible with lower-qubit priors or structural continuations;
5. they show at least weak support under dynamical metrics such as retention, leakage, purity, or drift.

For example, the pair

$$(157, 158)$$

may be represented in binary notation as

$$157 = 10011101_2,$$

and

$$158 = 10011110_2.$$

This binary representation is useful because the computational basis of an 8-qubit system is naturally indexed by 8-bit strings. However, the binary form alone does not prove that the pair is structurally important. The relevant question is whether the pair or its neighbourhood repeatedly behaves differently from the corresponding NULL structure.

A local hotspot neighbourhood around an anchor pair may be written as

$$\mathcal{N}_8(157, 158) \subset \mathcal{H}_8.$$

The associated evidence should then be evaluated by comparing REAL and NULL metrics inside that neighbourhood:

$$\Delta M_{8,\mathcal{N}} = M_{\text{REAL}}(8, \mathcal{N}_8(157, 158)) - M_{\text{NULL}}(8, \mathcal{N}_8(157, 158)).$$

If the metric is retention, the preferred condition is

$$\Delta R_{8,\mathcal{N}} > 0.$$

If the metric is leakage, the preferred condition is

$$\Delta \Lambda_{8,\mathcal{N}} < 0.$$

If the metric is local purity, the preferred condition is

$$\Delta P_{8,\mathcal{N}} > 0.$$

If the metric is structure drift, the preferred condition is

$$\Delta D_{8,\mathcal{N}} < 0.$$

Anchor continuation concerns the question of whether a hotspot at one qubit number has a related structure at the next qubit number. If

$$a_{n,j}$$

is an anchor at n qubits, then a direct embedded descendant may be written as

$$E_n(a_{n,j})$$

inside $\mathcal{H}_{n+1}$. However, in practice, the continuation may be approximate rather than exact. The higher-qubit hotspot may be shifted, split into multiple nearby candidates, or concentrated into a smaller kernel.

For this reason, continuation is written as

$$a_{n,j} \longrightarrow \hat{a}_{n+1,j'},$$

where $\hat{a}_{n+1,j'}$ denotes a predicted or observed successor anchor at $(n+1)$ qubits.

A useful continuation score may combine index proximity, signature similarity, static separation, and dynamical behaviour:

$$C_{n \rightarrow n+1}(j, j') = C(a_{n,j}, \hat{a}_{n+1,j'}; \Delta M_{\text{static}}, \Delta R, \Delta \Lambda, \Delta P, \Delta D, B).$$

Here, $C_{n \rightarrow n+1}(j, j')$ is high when the candidate at $(n+1)$ qubits appears structurally related to the candidate at n qubits and also performs favourably under REAL – NULL comparison.

This gives the following interpretation of hotspot continuation:

$$\text{anchor at } n \implies \text{predicted anchor neighbourhood at } (n+1).$$

The scaling programme from 4 to 8 qubits can therefore be understood as a search for recurrent anchor structures through the chain

$$a_4 \rightarrow a_5 \rightarrow a_6 \rightarrow a_7 \rightarrow a_8.$$

At early qubit numbers, the anchors may be broad and weakly localised. As the Hilbert-space dimension increases, the hypothesis is that some of these broad structures may concentrate into more specific hotspots or dynamic kernels. In this interpretation, an 8-qubit hotspot such as the region around (157,158) is most interesting if it can be viewed not as an isolated discovery, but as the endpoint or refinement of a lower-dimensional structural progression.

The role of anchor points is therefore methodological. They help transform the project from a purely retrospective scan into a predictive search. Once an anchor or hotspot is found at n qubits, it can be used as a prior for where to search at $(n+1)$ qubits. This is the first step toward a table-like organisation of soft-space structures, where recurring anchors, neighbourhoods, and kernels can be compared across qubit number, noise model, and dynamical metric.

4.5 Stepwise progression from 4 to 8 qubits

The scaling programme from 4 to 8 qubits is not treated as a single jump from dimension 16 to dimension 256. Instead, it is treated as a stepwise progression,

$$4 \rightarrow 5 \rightarrow 6 \rightarrow 7 \rightarrow 8,$$

where each transition doubles the Hilbert-space dimension and creates a new search problem. The purpose of the stepwise formulation is to evaluate whether structures observed at one qubit number can be used as priors for the next.

Let

$$\mathcal{S}_{n,j} \subset \mathcal{H}_n$$

denote a candidate soft-space region at n qubits. A successful continuation from n to $(n+1)$ qubits may be written schematically as

$$\mathcal{S}_{n,j} \longrightarrow \hat{\mathcal{S}}_{n+1,j'},$$

where $\hat{\mathcal{S}}_{n+1,j'}$ denotes a predicted or observed successor region at the next qubit number.

The continuation is evaluated using a combination of static and dynamical evidence. Static evidence asks whether the successor region remains separated from its matched NULL control. Dynamical evidence asks whether the successor region retains probability mass, leaks less, preserves local purity, or drifts less under Lindblad-type evolution.

A generic stepwise continuation score may be written as

$$C_{n \rightarrow n+1}(j, j') = C(\mathcal{S}_{n,j}, \hat{\mathcal{S}}_{n+1,j'}),$$

where $C_{n \rightarrow n+1}(j, j')$ increases when the candidate successor at $(n+1)$ qubits is structurally and dynamically compatible with the candidate at n qubits.

More explicitly, one may write

$$\begin{aligned} C_{n \rightarrow n+1}(j, j') = & w_1 A_{n \rightarrow n+1}(j, j') + w_2 \Delta M_{\text{static}}(n+1, j') \\ & + w_3 \overline{\Delta R}_{n+1, j'} - w_4 \overline{\Delta \Lambda}_{n+1, j'} \\ & + w_5 \overline{\Delta P}_{n+1, j'} - w_6 \overline{\Delta D}_{n+1, j'} \\ & + w_7 B_{n+1, j'}. \end{aligned}$$

Here, $A_{n \rightarrow n+1}(j, j')$ denotes anchor or signature compatibility between the lower-qubit candidate and the higher-qubit candidate. The term $\Delta M_{\text{static}}(n+1, j')$ denotes the static REAL – NULL contrast at the higher qubit number. The terms $\overline{\Delta R}_{n+1, j'}$, $\overline{\Delta \Lambda}_{n+1, j'}$, $\overline{\Delta P}_{n+1, j'}$, and $\overline{\Delta D}_{n+1, j'}$ denote time-aggregated contrasts in retention, leakage, purity, and drift. Finally, $B_{n+1, j'}$ denotes batch recurrence or stability.

The signs in this score reflect the preferred directions:

$$\overline{\Delta R}_{n+1, j'} > 0,$$

$$\overline{\Delta \Lambda}_{n+1, j'} < 0,$$

$$\overline{\Delta P}_{n+1, j'} > 0,$$

and

$$\overline{\Delta D}_{n+1, j'} < 0.$$

Thus, retention and purity contribute positively, while leakage and drift contribute favourably when they are smaller for REAL than for NULL.

From 4 to 5 qubits

The first transition is

$$\mathcal{H}_4 \rightarrow \mathcal{H}_5,$$

with dimensions

$$16 \rightarrow 32.$$

At 4 qubits, the Hilbert space is small enough for candidate structures to be inspected relatively directly. The main role of the 4-qubit stage is therefore to establish a baseline: which metrics separate REAL configurations from matched NULL controls, and whether any candidate subspaces recur across batches.

The transition to 5 qubits tests whether these baseline structures are merely artefacts of the smallest system size. A successful 4-to-5 continuation should show that at least some of the lower-qubit evidence survives the first dimensional doubling. The expected condition is not exact equality of structures, but preservation of statistical role:

$$\Delta M_{4, j} \neq 0 \implies \Delta M_{5, j'} \neq 0$$

for a structurally related successor candidate $\mathcal{S}_{5, j'}$.

At this stage, the most important evidence is static. The project should ask whether the same type of REAL – NULL separation remains visible after the Hilbert-space dimension increases from 16 to 32. Dynamical evidence may be evaluated, but it should be interpreted cautiously because the main purpose of this transition is to verify that the signal does not disappear immediately.

From 5 to 6 qubits

The second transition is

$$\mathcal{H}_5 \rightarrow \mathcal{H}_6,$$

with dimensions

$$32 \rightarrow 64.$$

At 6 qubits, the search space becomes large enough that recurrence and subspace selection become more important. If the soft-space hypothesis is meaningful, the evidence should begin to show not only that REAL and NULL differ, but that the difference is concentrated in selected regions.

The expected progression can be written as

$$\mathcal{S}_{5,j} \longrightarrow \widehat{\mathcal{S}}_{6,j'}.$$

A 6-qubit candidate becomes stronger if it satisfies both static and early dynamical criteria:

$$\Delta M_{\text{static}}(6, j') > 0,$$

and at least one favourable dynamical condition, such as

$$\overline{\Delta R}_{6,j'} > 0,$$

or

$$\overline{\Delta P}_{6,j'} > 0.$$

The 5-to-6 transition is therefore the first stage where the project can begin to distinguish between broadly static structure and potentially dynamically relevant structure. A candidate that is visible statically but collapses completely under Lindblad evolution is less compelling than a candidate that retains at least weak dynamical support.

From 6 to 7 qubits

The third transition is

$$\mathcal{H}_6 \rightarrow \mathcal{H}_7,$$

with dimensions

$$64 \rightarrow 128.$$

At 7 qubits, uniform scanning becomes increasingly inefficient. The number of possible candidate regions, neighbourhoods, and signature families is large enough that prior information from lower qubit numbers becomes methodologically important.

The expected progression is

$$\mathcal{S}_{6,j} \longrightarrow \widehat{\mathcal{S}}_{7,j'}.$$

The main question is whether the higher-dimensional structure begins to concentrate. If the lower-qubit signal is meaningful, one may expect that not all regions of $\mathcal{H}_7$ are equally informative. Instead, the signal should increasingly appear in selected anchor neighbourhoods or recurrent families.

This can be expressed as a refinement relation:

$$\widehat{\mathcal{S}}_{7,j'} \supset \mathcal{K}_{7,j'},$$

where $\mathcal{K}_{7,j'}$ denotes a smaller internal region or proto-kernel with stronger dynamical properties than the broader candidate subspace.

The 6-to-7 transition therefore tests whether the project can move from broad subspace identification toward kernel localisation. A favourable result is not merely that REAL-NUL separation persists, but that the separation becomes more localised, recurrent, and dynamically interpretable.

From 7 to 8 qubits

The fourth transition is

$$\mathcal{H}_7 \rightarrow \mathcal{H}_8,$$

with dimensions

$$128 \rightarrow 256.$$

This is the most important transition in the present 4-to-8 qubit study. At 8 qubits, the Hilbert space is large enough for anchor-like structures and recurrent hotspots to become more informative. Candidate regions such as neighbourhoods around

$$(157, 158)$$

become important if they appear not as isolated numerical accidents, but as recurrent structural coordinates.

The expected progression is

$$\mathcal{S}_{7,j} \longrightarrow \widehat{\mathcal{S}}_{8,j'}.$$

At this stage, the question is no longer simply whether REAL-NUL separation exists. The question is whether the separation has become concentrated enough to support targeted dynamic-kernel search.

A favourable 8-qubit candidate should satisfy several conditions:

1. it should be REAL-favoured under one or more static metrics;
2. it should recur across batches or repeated runs;
3. it should belong to a coherent anchor neighbourhood;
4. it should be compatible with lower-qubit priors or continuation rules;
5. it should show at least weak favourable behaviour under Lindblad-type metrics.

The strongest form of evidence would be a relation of the form

$$\widehat{\mathcal{S}}_{8,j'} \supset \mathcal{K}_{8,j'},$$

where $\mathcal{K}_{8,j'}$ is a dynamically favourable kernel inside the broader 8-qubit candidate region.

In this interpretation, an 8-qubit hotspot is not the end of the analysis. It is the beginning of a more focused search. The role of the 8-qubit stage is to identify candidate anchor neighbourhoods where later versions can zoom in, test sub-subspaces, and determine whether the observed structure has genuine dynamical persistence.

Summary of the stepwise progression

The stepwise scaling logic can be summarised as

$$\mathcal{S}_4 \rightarrow \mathcal{S}_5 \rightarrow \mathcal{S}_6 \rightarrow \mathcal{S}_7 \rightarrow \mathcal{S}_8,$$

with increasing emphasis on localisation:

$$\mathcal{S}_n \supset \mathcal{K}_n.$$

At lower qubit numbers, the main task is to establish that REAL-NULL separation exists. At higher qubit numbers, the main task is to determine where that separation concentrates and whether the concentrated regions possess favourable dynamical properties.

This progression motivates the later subspace-selective strategy. Rather than spending equal computational effort across all candidate regions, the lower-qubit results should guide the higher-qubit search. In this way, the 4-to-8 qubit track becomes a controlled attempt to move from detection to prediction:

$$\text{detect structure at } n \implies \text{predict where to search at } (n+1).$$

4.6 Toward a predictive soft-space scaling rule

The preceding subsections define the main ingredients needed for a predictive scaling rule: Lindblad evolution for an n -qubit system, embedding from n to $(n+1)$ qubits, dynamical metrics, anchor continuation, and stepwise progression from 4 to 8 qubits. The purpose of this final subsection is to combine these elements into a preliminary rule for predicting where candidate soft-space structures may appear as the Hilbert-space dimension increases.

The aim is not to claim a universal law at this stage. The present evidence is not yet sufficient for that. Rather, the goal is to formulate an operational scaling rule that can be tested, refined, and falsified in later versions of the project.

Let

$$\mathcal{S}_{n,j} \subset \mathcal{H}_n$$

denote the j -th candidate soft-space region at n qubits. The central predictive question is whether this candidate has a structurally related successor

$$\hat{\mathcal{S}}_{n+1,j'} \subset \mathcal{H}_{n+1}.$$

A general predictive rule may be written as

$$\hat{\mathcal{S}}_{n+1,j'} = \Phi_n(\mathcal{S}_{n,j}, H_n, \{L_{k,n}\}, A_{n,j}, \Delta M_{\text{static}}(n, j), \overline{\Delta R}_{n,j}, \overline{\Delta \Lambda}_{n,j}, \overline{\Delta P}_{n,j}, \overline{\Delta D}_{n,j}, B_{n,j}).$$

Here, Φ_n denotes an empirical transition rule from n to $(n+1)$ qubits. The inputs to the rule are:

1. the candidate subspace $\mathcal{S}_{n,j}$;
2. the effective Hamiltonian H_n ;
3. the Lindblad operators $\{L_{k,n}\}$;
4. an anchor or signature descriptor $A_{n,j}$;
5. the static REAL – NULL contrast $\Delta M_{\text{static}}(n, j)$;

6. the time-averaged retention contrast $\overline{\Delta R}_{n,j}$;
7. the time-averaged leakage contrast $\overline{\Delta \Lambda}_{n,j}$;
8. the time-averaged purity contrast $\overline{\Delta P}_{n,j}$;
9. the time-averaged drift contrast $\overline{\Delta D}_{n,j}$;
10. the batch-stability or recurrence score $B_{n,j}$.

This rule expresses the central methodological claim of the scaling programme: lower-qubit evidence should not merely be reported retrospectively; it should be used to generate higher-qubit search priors.

A candidate at $(n+1)$ qubits is therefore not evaluated in isolation. It is evaluated partly by asking whether it is compatible with the evidence already observed at n qubits. In this way, the project moves from independent scans at each qubit number toward a structured continuation analysis.

A scalar score for a predicted successor may be written as

$$\begin{aligned} Q_{n+1,j'} = & \alpha \Delta M_{\text{static}}(n+1, j') + \beta \overline{\Delta R}_{n+1,j'} \\ & - \gamma \overline{\Delta \Lambda}_{n+1,j'} + \delta \overline{\Delta P}_{n+1,j'} \\ & - \epsilon \overline{\Delta D}_{n+1,j'} + \eta C_{n \rightarrow n+1}(j, j') \\ & + \mu B_{n+1,j'}. \end{aligned}$$

Here, $Q_{n+1,j'}$ is the total score assigned to candidate successor $\hat{\mathcal{S}}_{n+1,j'}$. The term

$$C_{n \rightarrow n+1}(j, j')$$

measures continuation compatibility between $\mathcal{S}_{n,j}$ and $\hat{\mathcal{S}}_{n+1,j'}$. The weights

$$\alpha, \beta, \gamma, \delta, \epsilon, \eta, \mu$$

control the relative importance of static separation, retention, leakage, purity, drift, continuation compatibility, and batch recurrence.

The signs in the score encode the preferred directions. Static separation, retention, purity, continuation compatibility, and recurrence increase the score when they are favourable. Leakage and drift enter with negative signs because lower leakage and lower drift are preferred.

The preferred qualitative pattern is therefore

$$\Delta M_{\text{static}} > 0,$$

$$\overline{\Delta R} > 0,$$

$$\overline{\Delta \Lambda} < 0,$$

$$\overline{\Delta P} > 0,$$

$$\overline{\Delta D} < 0,$$

$$C_{n \rightarrow n+1} \text{ high,}$$

and

B high.

A candidate with this pattern is a strong successor candidate. A candidate that satisfies only the static terms is weaker but may still be useful as a search prior. A candidate that satisfies neither static nor dynamical criteria should not be treated as evidence for soft-space continuation.

This suggests a practical decision rule:

$\mathcal{S}_{n+1,j'}$ is accepted as a successor candidate

if

$$Q_{n+1,j'} > \theta_{n+1},$$

where θ_{n+1} is a qubit-dependent threshold. The threshold should not be chosen arbitrarily. It should be calibrated against the distribution of scores obtained from matched NULL controls, repeated batches, and alternative noise settings.

Equivalently, the threshold may be expressed in terms of a standardised score,

$$Z_{n+1,j'} = \frac{Q_{n+1,j'} - \mu_{\text{NULL}}(n+1)}{\sigma_{\text{NULL}}(n+1)},$$

where $\mu_{\text{NULL}}(n+1)$ and $\sigma_{\text{NULL}}(n+1)$ are the mean and standard deviation of the corresponding NULL score distribution. A candidate becomes more credible when

$$Z_{n+1,j'}$$

is large relative to the NULL background.

This formulation also clarifies the role of 8-qubit hotspots. A recurrent anchor neighbourhood such as the region around

$$(157, 158)$$

should not be treated as an isolated discovery. It should be scored according to the same rule:

$$Q_{8,\mathcal{N}(157,158)} = \alpha \Delta M_{\text{static}} + \beta \overline{\Delta R} - \gamma \overline{\Delta \Lambda} + \delta \overline{\Delta P} - \epsilon \overline{\Delta D} + \eta C_{7 \rightarrow 8} + \mu B_8.$$

The neighbourhood becomes a stronger candidate if it combines static REAL–NULL separation, recurrence across batches, compatibility with lower-qubit priors, and favourable Lindblad-based dynamical behaviour.

The long-term ambition of this scaling rule is to build a table-like organisation of soft-space structures. This would not be a periodic table in the chemical sense. Rather, it would be an empirical classification of recurring subspace patterns, anchors, and dynamic kernels across qubit number:

Qubits	Candidate anchor	Static evidence	Dynamical evidence
4	a_4	$\Delta M_{\text{static}}(4)$	$\overline{\Delta M}_{\text{dyn}}(4)$
5	a_5	$\Delta M_{\text{static}}(5)$	$\overline{\Delta M}_{\text{dyn}}(5)$
6	a_6	$\Delta M_{\text{static}}(6)$	$\overline{\Delta M}_{\text{dyn}}(6)$
7	a_7	$\Delta M_{\text{static}}(7)$	$\overline{\Delta M}_{\text{dyn}}(7)$
8	a_8	$\Delta M_{\text{static}}(8)$	$\overline{\Delta M}_{\text{dyn}}(8)$

Such a table would allow the project to ask whether anchor structures are isolated, recurrent, transformed, or refined as n increases. It would also make the predictive character of the framework explicit: the result at one qubit number becomes a hypothesis for the next.

The current status of the project supports a cautious version of this programme. Static REAL – NULL separation is the strongest component of the evidence. Dynamical evidence is weaker, but it becomes important when it aligns with static separation, recurrence, and anchor continuation. The predictive scaling rule should therefore be interpreted as a framework for the next phase, not as a final law.

In this framework, the 4-to-8 qubit track has a specific methodological role. It provides the first controlled sequence in which static separation, Lindblad-based dynamical testing, anchor recurrence, and continuation analysis can be studied together. The purpose of later versions, including v24_8 and beyond, is to refine this rule by testing whether dynamically favourable kernels can be predicted from lower-qubit evidence rather than discovered only retrospectively.

5 Methodology

5.1 Overview of the experimental pipeline

For each qubit number

$$n \in \{4, 5, 6, 7, 8\},$$

the analysis follows a common experimental pipeline. The purpose of using the same broad pipeline across qubit numbers is to make the 4-to-8 qubit sequence comparable. Each run is treated as part of a controlled scaling study rather than as an isolated numerical experiment.

The pipeline consists of the following stages:

- (1) define the qubit number n and corresponding Hilbert space $\mathcal{H}_n$;
- (2) generate or load the REAL configurations;
- (3) generate matched NULL controls;
- (4) compute static metrics for both REAL and NULL systems;
- (5) identify candidate subspaces, signature families, anchor regions, or hotspots;
- (6) evaluate batch stability and recurrence across independent seeds or batches;
- (7) apply open-system dynamics to selected candidates;
- (8) compute dynamical metrics such as retention, leakage, purity, and drift;
- (9) compare REAL and NULL results under both static and dynamical metrics;
- (10) summarise the strongest candidate regions for later continuation analysis.

This pipeline is designed to separate three different evidential layers. The first layer is static separation: whether REAL systems differ from matched NULL controls before open-system evolution. The second layer is stability: whether the same type of structure recurs across batches, seeds, or related signature families. The third layer is dynamical behaviour: whether the candidate structures retain favourable properties under Lindblad-type evolution.

The general form of the evidence is therefore

$$\text{candidate} \implies \text{static separation} \implies \text{batch recurrence} \implies \text{dynamical testing}.$$

A candidate soft-space region is strongest when all three evidential layers point in the same direction.

5.2 Qubit range

The analysed qubit range is

$$n = 4, 5, 6, 7, 8.$$

The corresponding Hilbert-space dimensions are

$$\dim(\mathcal{H}_n) = 2^n.$$

The dimensions used in the study are shown in Table 1.

Table 1: Hilbert-space dimensions for the analysed qubit systems.

Number of qubits n	Hilbert-space dimension 2^n
4	16
5	32
6	64
7	128
8	256

The transition from 4 to 8 qubits therefore corresponds to the dimensional progression

$$16 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256.$$

This range is large enough to test whether candidate subspace structure persists beyond the smallest system sizes, but still small enough to allow detailed numerical inspection, repeated batch testing, and controlled REAL-NULL comparisons.

The purpose of the qubit sweep is not to demonstrate large-scale quantum advantage. Rather, it is to evaluate whether low-qubit Hilbert spaces already contain reproducible internal structure, and whether such structure can be followed as the dimension increases.

5.3 Construction of REAL and NULL comparisons

The central comparison is between REAL configurations and matched NULL controls. For a metric M , define the basic contrast as

$$\Delta M = M_{\text{REAL}} - M_{\text{NULL}}.$$

When the metric depends on qubit number n and candidate region j , the contrast is written as

$$\Delta M_{n,j} = M_{\text{REAL}}(n, j) - M_{\text{NULL}}(n, j).$$

The role of the NULL system is to provide a matched reference model. It should preserve selected baseline properties while removing or randomising the structure being tested. This is important because a raw score in a high-dimensional Hilbert space is not sufficient evidence of meaningful structure. A high score may arise from dimensionality, sampling, spectral properties, or numerical artefacts. The relevant question is whether the same score remains unusual relative to a matched control.

Thus, a candidate region is interpreted as meaningful only if its REAL-NULL contrast is stable, directionally interpretable, and reproducible across repeated runs.

5.4 Static metrics

The static analysis compares REAL and NULL systems before open-system evolution. A generic static difference is written as

$$\Delta M_{\text{static}}(n, j) = M_{\text{static,REAL}}(n, j) - M_{\text{static,NULL}}(n, j).$$

Static metrics are used to identify candidate regions that differ from their matched controls at baseline. These metrics may include entropy-related structure, ranked score separation, overlap structure, signature-family enrichment, and recurrence across batches.

An entropy-related contrast may be written schematically as

$$\Delta S_{n,j} = S_{\text{REAL}}(n, j) - S_{\text{NULL}}(n, j),$$

where S denotes the entropy measure used in the corresponding implementation. The interpretation of the sign depends on the specific entropy definition and the construction of the candidate region. For this reason, entropy-related quantities should always be reported with their operational definition.

A ranked separation score may be written as

$$\Delta Q_{n,j} = Q_{\text{REAL}}(n, j) - Q_{\text{NULL}}(n, j),$$

where Q denotes a ranking or scoring function assigned to candidate regions. A positive value indicates that the REAL candidate ranks above its matched NULL counterpart under the chosen score.

Signature-family enrichment can be represented by a recurrence or enrichment count. If a signature family F appears $c_{\text{REAL}}(F)$ times in REAL candidates and $c_{\text{NULL}}(F)$ times in NULL candidates, an enrichment contrast may be written as

$$\Delta c(F) = c_{\text{REAL}}(F) - c_{\text{NULL}}(F).$$

Alternatively, a ratio form may be used,

$$E(F) = \frac{c_{\text{REAL}}(F) + \epsilon}{c_{\text{NULL}}(F) + \epsilon},$$

where $\epsilon > 0$ is a small regularisation constant used to avoid division by zero.

Overlap or fidelity-based static comparison may be written generically as

$$F(\psi, \phi) = |\langle \psi | \phi \rangle|^2$$

for pure states, or

$$F(\rho, \sigma) = \left(\text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right)^2$$

for density matrices. Such quantities can be used to compare candidate states, subspaces, or local neighbourhoods.

The static metrics serve as the first filter in the methodology. A candidate region that does not separate statically from its matched NULL control is not treated as strong evidence for a soft space. Conversely, a region with stable static separation becomes a candidate for later dynamical testing.

5.5 Dynamical metrics

Dynamical metrics are computed after selected candidate regions have been exposed to open-system evolution. The state evolution is modelled using Lindblad-type dynamics,

$$\frac{d\rho_n(t)}{dt} = -i[H_n, \rho_n(t)] + \sum_k \left(L_{k,n} \rho_n(t) L_{k,n}^\dagger - \frac{1}{2} \{ L_{k,n}^\dagger L_{k,n}, \rho_n(t) \} \right).$$

The specific Lindblad operators $L_{k,n}$ depend on the chosen noise model. In the present project, the relevant family of tests includes dephasing-style and amplitude-damping-style channels. These channels are used to stress-test candidate subspaces under different forms of open-system disturbance.

For a candidate subspace $\mathcal{S}_{n,j}$ with projector $P_{n,j}$, retention is defined as

$$R_{n,j}(t) = \text{Tr}(P_{n,j} \rho_n(t)).$$

Leakage is defined as

$$\Lambda_{n,j}(t) = 1 - R_{n,j}(t).$$

The local purity inside a projected and renormalised candidate region is

$$P_{n,j}(t) = \text{Tr} \left[\left(\rho_{n,j}^{\text{proj}}(t) \right)^2 \right],$$

where

$$\rho_{n,j}^{\text{proj}}(t) = \frac{P_{n,j} \rho_n(t) P_{n,j}}{\text{Tr}(P_{n,j} \rho_n(t))}.$$

Structure drift is represented generically as

$$D_{n,j}(t) = d(S_{n,j}(0), S_{n,j}(t)),$$

where $S_{n,j}(t)$ is a structural descriptor and d is an implementation-specific distance function. The corresponding REAL-NULI contrasts are

$$\Delta R_{n,j}(t) = R_{\text{REAL}}(n, j, t) - R_{\text{NULL}}(n, j, t),$$

$$\Delta \Lambda_{n,j}(t) = \Lambda_{\text{REAL}}(n, j, t) - \Lambda_{\text{NULL}}(n, j, t),$$

$$\Delta P_{n,j}(t) = P_{\text{REAL}}(n, j, t) - P_{\text{NULL}}(n, j, t),$$

and

$$\Delta D_{n,j}(t) = D_{\text{REAL}}(n, j, t) - D_{\text{NULL}}(n, j, t).$$

The preferred directions are

$$\Delta R_{n,j}(t) > 0,$$

$$\Delta \Lambda_{n,j}(t) < 0,$$

$$\Delta P_{n,j}(t) > 0,$$

and

$$\Delta D_{n,j}(t) < 0.$$

For time-aggregated analysis, a generic dynamical metric $X_{n,j}(t)$ may be summarised by

$$\bar{X}_{n,j} = \frac{1}{T} \int_0^T X_{n,j}(t) dt.$$

In numerical implementations, this is approximated by

$$\bar{X}_{n,j} \approx \frac{1}{m} \sum_{\ell=1}^m X_{n,j}(t_\ell).$$

Area-under-curve quantities may also be used, for example

$$\text{AUC}(R_{n,j}) = \int_0^T R_{n,j}(t) dt.$$

The dynamical signal is interpreted more cautiously than the static signal. The current evidence trail indicates that static REAL-NULL separation is stronger and more stable, while dynamical separation is weaker but directionally suggestive in selected cases. Dynamical metrics therefore serve as a second filter rather than the sole basis for accepting a candidate region.

5.6 Batch stability and recurrence

A candidate subspace is not considered robust merely because it performs well in a single run. Stability across batches, seeds, or repeated configurations is required. If there are B batches, a simple recurrence score for candidate region $\mathcal{S}_{n,j}$ can be written as

$$\text{Recurrence}(\mathcal{S}_{n,j}) = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{\mathcal{S}_{n,j} \text{ appears in batch } b\}.$$

A recurrence value close to 1 indicates that the candidate appears in most batches. A value close to 0 indicates that the candidate is not stable across repeated runs.

For directional metrics, batch stability also requires sign consistency. If $\Delta M_{n,j}^{(b)}$ denotes the REAL-NULL contrast in batch b , then a simple sign-stability score may be written as

$$\text{SignStability}(M) = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\left\{\text{sign}\left(\Delta M_{n,j}^{(b)}\right) = s^*\right\},$$

where s^* is the preferred or observed dominant sign.

In practical terms, a signal is more credible if it appears with the same sign in multiple independent batches and if the same or neighbouring signature families recur across seeds. For example, a candidate observed in only one batch is treated as provisional, while a candidate recurring in two or three out of three batches is substantially more credible.

Batch stability is particularly important for anchor points and hotspots. An index pair or neighbourhood such as

$$(157, 158)$$

is not meaningful merely because it appears once with a high score. It becomes meaningful only if the associated neighbourhood shows recurring REAL-favoured behaviour, preferably under both static and dynamical metrics.

5.7 Candidate selection and evidence hierarchy

The methodology uses an evidence hierarchy. Candidate regions are first screened by static metrics. They are then filtered by recurrence and batch stability. Finally, selected candidates are tested dynamically under Lindblad-type evolution.

This hierarchy can be written schematically as

$$\mathcal{S}_{n,j} \xrightarrow{\text{static}} \mathcal{S}_{n,j}^* \xrightarrow{\text{recurrence}} \mathcal{S}_{n,j}^{**} \xrightarrow{\text{dynamic}} \mathcal{K}_{n,j}.$$

Here, $\mathcal{S}_{n,j}$ is an initial candidate, $\mathcal{S}_{n,j}^*$ is a statically favoured candidate, $\mathcal{S}_{n,j}^{**}$ is a recurrent candidate, and $\mathcal{K}_{n,j}$ denotes a possible dynamically favourable kernel.

This hierarchy reflects the current interpretation of the project. Static separation provides the strongest evidence. Recurrence protects against one-off numerical artefacts. Dynamical testing evaluates whether the candidate region retains structure under noise. A region that survives all three stages is the strongest candidate for later subspace-selective analysis.

6 Version History and Experimental Development

The project developed through a sequence of experimental versions. The purpose of this section is not to provide a chronological diary of every run, but to summarise the main methodological transitions. Each stage introduced a specific improvement in the evidential framework: first the establishment of REAL-NULL separation, then scaling across qubit number, then open-system testing, and finally targeted subspace-selective search.

The development can be summarised as a transition through four methodological phases:

$$\text{baseline} \implies \text{scaling} \implies \text{dynamical testing} \implies \text{subspace selective refinement}.$$

Earlier versions focused on whether any reproducible REAL-NULL signal existed. Later versions increasingly focused on where the signal was concentrated and whether those regions could survive Lindblad-type dynamical stress tests.

6.1 Early 4-qubit baseline

The 4-qubit experiments served as the baseline for the project. At this stage,

$$\dim(\mathcal{H}_4) = 2^4 = 16.$$

This dimension is small enough to allow relatively direct inspection of candidate structures, while still being large enough to contain non-trivial internal subspace variation.

The primary goal of the 4-qubit baseline was to determine whether REAL configurations and matched NULL controls behaved identically under the chosen metrics. The relevant question was not whether the 4-qubit system was computationally large, but whether it already contained detectable internal structure.

The earliest evidence came from static REAL-NULL separation. In generic form, this was measured by contrasts such as

$$\Delta M_{4,j} = M_{\text{REAL}}(4, j) - M_{\text{NULL}}(4, j),$$

where j denotes a candidate region, family, or signature. The important observation was that selected metrics did not behave like random fluctuations around a matched control. Instead, some candidate regions showed directionally interpretable differences.

At this stage, the main contribution was methodological. The 4-qubit experiments established the basic comparison principle:

raw metric value is less important than REAL – NULL contrast.

This principle became the foundation for all later versions.

6.2 Extension to 5 and 6 qubits

The next stage extended the analysis to 5 and 6 qubits. The Hilbert-space dimensions increased as

$$\dim(\mathcal{H}_5) = 32$$

and

$$\dim(\mathcal{H}_6) = 64.$$

The purpose of this stage was to test whether the early REAL-NULL signal was specific to the smallest system size, or whether it persisted when the Hilbert space expanded.

The methodological question changed from

does any separation exist?

to

does separation persist under dimensional scaling?

At 5 and 6 qubits, the number of possible subspaces, signature families, and candidate regions increases substantially. This makes isolated high scores less convincing. A signal becomes more meaningful if it remains directionally stable, recurs across batches, or appears in structurally related regions.

The 5- and 6-qubit stages therefore introduced a stronger emphasis on recurrence and stability. A candidate structure became more credible if it satisfied conditions such as

$$\Delta M_{n,j}^{(b)} \text{ has the same preferred sign across batches}$$

for

$$n \in \{5, 6\}.$$

This stage also began to expose the need for a more systematic treatment of candidate selection. As the Hilbert-space dimension increased, it became less useful to treat all regions as equally informative. The results suggested that later versions should focus more directly on regions where the REAL-NULL contrast was largest or most stable.

6.3 Extension to 7 and 8 qubits

The extension to 7 and 8 qubits increased the Hilbert-space dimensions to

$$\dim(\mathcal{H}_7) = 128$$

and

$$\dim(\mathcal{H}_8) = 256.$$

This stage was important because the search problem became qualitatively different. At 8 qubits, the space is still numerically accessible, but large enough that uniform inspection

becomes increasingly inefficient. The project therefore began to shift from broad scanning toward prioritised subspace analysis.

The central question became

where is the signal concentrated?

rather than merely

does the signal exist?

This distinction is important. If REAL-NULL separation is distributed uniformly across the entire Hilbert space, then there may be little reason to focus on specific soft-space regions. If, however, the separation concentrates in selected anchor neighbourhoods, recurrent families, or hotspot regions, then the project can become predictive: lower-qubit evidence can guide higher-qubit search.

The 8-qubit stage is therefore where anchor-based thinking becomes especially relevant. Candidate regions such as neighbourhoods around recurrent pairs, for example

$$(157, 158),$$

are not interpreted as meaningful simply because of their numerical labels. Their importance depends on recurrence, REAL-favoured separation, local neighbourhood coherence, and compatibility with lower-qubit priors.

This stage also motivated the transition toward v24_8-style methods. The core methodological lesson was that the larger Hilbert space should not be searched uniformly. Instead, the analysis should prioritise subspaces where previous evidence suggests that structure is likely to concentrate.

6.4 Introduction of open-system and Lindblad-type testing

After static REAL-NULL separation had been established as the strongest evidence layer, later versions introduced open-system tests. These tests used Lindblad-type dynamics to evaluate whether candidate regions retained favourable behaviour under noise, dissipation, or decoherence.

The dynamical question was stronger than the static question. Static separation asks whether a region differs from its matched control at baseline. Dynamical testing asks whether that region retains structure over time.

For a candidate region $\mathcal{S}_{n,j}$, the relevant dynamical quantities included retention,

$$R_{n,j}(t),$$

leakage,

$$\Lambda_{n,j}(t),$$

local purity,

$$P_{n,j}(t),$$

and structure drift,

$$D_{n,j}(t).$$

The preferred pattern was

$$\Delta R_{n,j}(t) > 0,$$

$$\Delta\Lambda_{n,j}(t) < 0,$$

$$\Delta P_{n,j}(t) > 0,$$

and

$$\Delta D_{n,j}(t) < 0.$$

The empirical lesson from these versions was that dynamical evidence is more difficult to obtain than static evidence. The static signal was stronger and more reproducible. The dynamical signal was weaker, but in selected cases directionally consistent. This led to an important refinement of the project logic: Lindblad dynamics should be treated as a second evidential filter, not as the first screening mechanism.

6.5 Transition toward subspace-selective methods

The later methodological transition was from global detection to subspace-selective refinement. Earlier versions asked:

does REAL-NULL separation exist?

Later versions increasingly asked:

where is the separation strongest, most stable, and most dynamically favourable?

This is the intellectual transition from static discovery to dynamic-kernel identification. A broad candidate region may show static separation, but only a smaller internal part of that region may show favourable retention, low leakage, high local purity, or low drift. This motivates a refinement relation of the form

$$\mathcal{S}_{n,j} \supset \mathcal{K}_{n,j},$$

where $\mathcal{S}_{n,j}$ is a broader candidate soft-space region and $\mathcal{K}_{n,j}$ is a smaller candidate dynamic kernel.

This transition is central to the later v24_8 and v24_14-style direction of the project. The method no longer treats all candidate regions equally. Instead, it uses static separation, recurrence, anchor continuation, and dynamical metrics to decide where computational effort should be concentrated.

The resulting development path can be summarised as

$$\begin{aligned} \text{static separation} &\implies \text{recurrent candidates} \\ &\implies \text{anchor neighbourhoods} \\ &\implies \text{dynamic kernels.} \end{aligned}$$

This section therefore frames the experimental history as a methodological progression. The project began by detecting that REAL and NULL systems were not behaving identically. It then scaled this comparison across qubit number, introduced dynamical testing, and finally moved toward targeted search for the subspaces where the signal is most concentrated.

7 Results

7.1 Static REAL-NULL separation

Across the 4-to-8 qubit track, the most robust finding is static separation between REAL configurations and spectrum-matched NULL controls. This separation appears before open-system evolution is applied and therefore serves as the primary evidence layer in the present study.

For a static metric M_{static} , the relevant contrast is

$$\Delta M_{\text{static}}(n, j) = M_{\text{static,REAL}}(n, j) - M_{\text{static,NULL}}(n, j),$$

where n is the qubit number and j denotes a candidate subspace, signature family, or anchor neighbourhood.

The strongest overall pattern is that selected REAL regions separate from their matched NULL controls more consistently than expected under a purely random or structure-free interpretation. This does not imply that every candidate region is meaningful. Rather, it indicates that the REAL-favoured signal is concentrated in selected regions and families.

The static results should be summarised in terms of three quantities:

- (1) the direction of the REAL-NULL contrast;
- (2) the magnitude of the contrast;
- (3) the stability of the contrast across batches or repeated runs.

A compact summary table is shown in Table 2. The entries should be replaced by the exact values from the corresponding run outputs.

Table 2: Static REAL-NULL separation across qubit sizes. Placeholder entries should be replaced with measured results from the versioned runs.

Qubits	Main static metric	REAL-NULL direction	Batch stability
4	[insert metric]	[positive/negative/mixed]	[insert stability]
5	[insert metric]	[positive/negative/mixed]	[insert stability]
6	[insert metric]	[positive/negative/mixed]	[insert stability]
7	[insert metric]	[positive/negative/mixed]	[insert stability]
8	[insert metric]	[positive/negative/mixed]	[insert stability]

The intended interpretation of Table 2 is not that the same numerical value should appear at every qubit number. The relevant question is whether the same type of separation persists as the Hilbert-space dimension increases:

$$16 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256.$$

The current evidence trail supports the view that static separation is the strongest and most reproducible component of the project. It provides the main basis for identifying candidate soft-space regions and for selecting regions to be tested dynamically.

7.2 Dynamical signal under open-system evolution

The dynamical results are more subtle than the static results. Under Lindblad-type open-system evolution, selected REAL-favoured regions show indications of improved retention, reduced leakage, higher local purity, or lower structural drift compared with matched NULL controls. However, the magnitude of the dynamical signal is weaker than the static separation.

For a dynamical metric M_{dyn} , the time-dependent contrast is

$$\Delta M_{\text{dyn}}(n, j, t) = M_{\text{dyn,REAL}}(n, j, t) - M_{\text{dyn,NULL}}(n, j, t).$$

For time-aggregated comparison, this may be summarised as

$$\overline{\Delta M}_{\text{dyn}}(n, j) = \frac{1}{T} \int_0^T \Delta M_{\text{dyn}}(n, j, t) dt.$$

The principal dynamical metrics are retention, leakage, local purity, and structure drift. Their preferred directions are

$$\Delta R_{n,j} > 0,$$

$$\Delta \Lambda_{n,j} < 0,$$

$$\Delta P_{n,j} > 0,$$

and

$$\Delta D_{n,j} < 0.$$

Thus, a dynamically favourable REAL region should retain more probability mass, leak less, maintain higher local purity, and drift less than the matched NULL region.

The dynamical evidence should be reported cautiously. A weak dynamical signal does not by itself establish a robust dynamic soft space. However, it becomes meaningful when it aligns with static separation and recurrence. In that case, Lindblad-type evolution acts as a second filter: it helps identify which statically favoured regions are also dynamically promising.

A compact summary table for the dynamical results is shown in Table 3. Placeholder entries should be replaced with measured values from the relevant run outputs.

Table 3: Dynamical REAL-NULL comparison under open-system evolution. Placeholder entries should be replaced with measured results.

Qubits	Noise model	Metric	REAL-NULL effect	Interpretation
4	[insert]	[insert]	[insert]	[insert]
5	[insert]	[insert]	[insert]	[insert]
6	[insert]	[insert]	[insert]	[insert]
7	[insert]	[insert]	[insert]	[insert]
8	[insert]	[insert]	[insert]	[insert]

The present interpretation is therefore:

static signal: strong

and

dynamical signal: weaker but directionally informative.

This asymmetry is central to the paper. The dynamic results should not be overstated, but they are sufficiently suggestive to motivate targeted dynamic-kernel search in later versions.

7.3 Recurrent subspaces and signature families

A key observation is that some candidate regions recur across batches, runs, or versioned experiments. Such recurrence is more important than isolated high performance. A single high-scoring candidate may arise by chance, but a candidate that recurs across independent batches is more likely to reflect stable structure.

For a candidate region $\mathcal{S}_{n,j}$, recurrence can be written as

$$\text{Recurrence}(\mathcal{S}_{n,j}) = \frac{1}{B} \sum_{b=1}^B \mathbf{1} \{ \mathcal{S}_{n,j} \text{ appears in batch } b \}.$$

In the 8-qubit stage, recurrent anchor-like regions and index-pair structures are especially important because the Hilbert space has dimension

$$2^8 = 256.$$

At this size, a candidate such as a neighbourhood around

$$(157, 158)$$

should be interpreted only through its recurrence and metric behaviour. The numerical pair itself is not sufficient evidence. It becomes meaningful only if it appears as part of a coherent REAL-favoured neighbourhood, survives repeated testing, and aligns with static or dynamical metrics.

A candidate recurrent track should therefore be reported using the following structure:

- (1) the candidate anchor, index pair, or signature family;
- (2) the qubit number at which it appears;
- (3) the batches or repeats in which it appears;
- (4) the associated static REAL-NULL contrast;
- (5) any dynamical support from retention, leakage, purity, or drift;
- (6) whether the candidate appears compatible with lower-qubit priors.

A placeholder table for recurrent structures is shown in Table 4.

Table 4: Template for recurrent subspaces, anchors, or signature families. Replace placeholder entries with measured run results.

Qubits	Candidate region	Recurrence	Static support	Dynamical support
4	[insert]	[insert]	[insert]	[insert]
5	[insert]	[insert]	[insert]	[insert]
6	[insert]	[insert]	[insert]	[insert]
7	[insert]	[insert]	[insert]	[insert]
8	[insert, e.g. (157,158)]	[insert]	[insert]	[insert]

This table is important because it connects the numerical evidence to the scaling interpretation. If the same or related structures recur across qubit number, then the project moves beyond isolated detection and toward continuation analysis.

7.4 Evidence summary

The results can be summarised in four main points.

First, static REAL-NULL separation is the strongest and most reproducible signal. This is the primary empirical basis for candidate soft-space identification.

Second, dynamical separation under Lindblad-type evolution is weaker. It should not be treated as conclusive on its own. However, it remains important when the direction of the signal aligns with static separation, recurrence, and anchor continuation.

Third, selected candidate regions, signature families, or anchor neighbourhoods recur across batches or versions. This recurrence is essential because it distinguishes potentially stable structure from one-off numerical artefacts.

Fourth, the results motivate a transition from uniform scanning to targeted subspace-selective search. The next methodological step is not merely to repeat the same analysis at larger qubit numbers, but to use lower-qubit evidence to predict where dynamically promising regions may appear at higher qubit numbers.

The overall evidence hierarchy is therefore

$$\begin{aligned} \text{static separation} &\implies \text{recurrence} \\ &\implies \text{dynamical support} \\ &\implies \text{candidate dynamic kernel.} \end{aligned}$$

The present results support the first two layers most strongly and provide suggestive evidence for the third. This is sufficient to motivate the next stage of the project: hierarchical search for smaller dynamically favourable kernels inside statically favoured soft-space regions.

8 Interpretation

The results suggest that low-qubit Hilbert spaces may contain non-uniform internal structure that can be detected through matched REAL-NULL comparisons. The strongest evidence concerns static structure. The dynamical evidence is weaker, but in selected cases it is directionally consistent with the static results. This suggests that some statically favoured regions may also have limited dynamical significance under open-system evolution.

The interpretation should therefore be deliberately cautious. The results do not prove that the identified regions are fault-tolerant logical qubits, nor do they establish a new quantum architecture. Rather, they identify candidate subspaces where the measured behaviour differs from matched controls. In this sense, the project provides an empirical framework for detecting and prioritising structured regions inside low-qubit Hilbert spaces.

The central interpretation is

$$\begin{aligned} \text{static separation} &\implies \text{candidate soft space,} \\ \text{while} \end{aligned}$$

$$\text{static separation} + \text{dynamical support} \implies \text{candidate dynamic kernel.}$$

Thus, static evidence and dynamical evidence do not play the same role. Static evidence is the primary discovery mechanism. Dynamical evidence is a stricter filter that tests whether selected regions retain favourable behaviour under noise, dissipation, or decoherence.

8.1 Why static evidence is stronger

Static metrics are measured before open-system evolution is applied. They are therefore less exposed to noise-model sensitivity, time-discretisation choices, numerical integration effects, and

assumptions about the Lindblad operators. This makes static metrics a cleaner first layer of evidence.

A static contrast has the generic form

$$\Delta M_{\text{static}}(n, j) = M_{\text{static,REAL}}(n, j) - M_{\text{static,NULL}}(n, j).$$

If this contrast is stable across batches, then it provides evidence that the REAL region differs from its matched NULL counterpart before dynamical stress is applied.

The static signal is easier to interpret because it depends primarily on the structure of the candidate region and the construction of the matched control. By contrast, a dynamical signal also depends on the chosen Hamiltonian, the Lindblad operators, the time horizon, the time discretisation, and the selected dynamical summary statistic.

This does not mean that static evidence is sufficient by itself. A region may be statically unusual but dynamically fragile. However, static evidence is the most reliable entry point for identifying candidate regions. It tells the method where to look.

In this sense, static analysis serves as a screening stage:

$$\mathcal{H}_n \implies \{\mathcal{S}_{n,j}^*\},$$

where $\{\mathcal{S}_{n,j}^*\}$ denotes the set of statically favoured candidate subspaces selected for further testing.

8.2 Why the dynamical signal still matters

The dynamical signal is weaker than the static signal, but it remains important when it is directionally consistent. A small dynamical effect is not persuasive if it appears only once, changes sign across batches, or is disconnected from the static structure. It becomes more meaningful when it aligns with static separation, recurrence, and anchor continuation.

For a candidate region $\mathcal{S}_{n,j}$, the relevant dynamical contrasts include

$$\Delta R_{n,j}(t),$$

$$\Delta \Lambda_{n,j}(t),$$

$$\Delta P_{n,j}(t),$$

and

$$\Delta D_{n,j}(t).$$

The preferred pattern is

$$\Delta R_{n,j}(t) > 0,$$

$$\Delta \Lambda_{n,j}(t) < 0,$$

$$\Delta P_{n,j}(t) > 0,$$

and

$$\Delta D_{n,j}(t) < 0.$$

This pattern means that the REAL-favoured region retains more probability mass, leaks less, maintains higher local purity, and drifts less than the matched NULL control.

The present results should not be interpreted as showing that the dynamical signal is already strong enough to define a final class of robust soft spaces. A more appropriate interpretation is that the signal is weak but informative. It provides a rational basis for targeted search: if static metrics identify promising regions, and dynamical metrics show even modest support in the same regions, then later versions should zoom into those regions rather than scan the full Hilbert space uniformly.

Thus, the dynamical result matters because it changes the search strategy. It motivates a transition from broad detection to selective refinement:

$$\text{broad static candidate} \implies \text{dynamically tested candidate.}$$

8.3 From broad subspaces to dynamic kernels

A broad candidate subspace may contain smaller internal regions with stronger dynamical properties. This motivates the distinction between a soft-space candidate and a dynamic kernel.

Let

$$\mathcal{S}_{n,j} \subset \mathcal{H}_n$$

denote a broad candidate soft-space region. The hypothesis is that inside some such regions there may exist smaller subregions

$$\mathcal{K}_{n,j} \subset \mathcal{S}_{n,j},$$

where $\mathcal{K}_{n,j}$ has better dynamical behaviour than the broader region.

This refinement can be written as

$$\mathcal{S}_{n,j} \supset \mathcal{K}_{n,j}.$$

The broader region $\mathcal{S}_{n,j}$ may be identified by static separation, while the smaller kernel $\mathcal{K}_{n,j}$ should be identified by stronger dynamical criteria, such as improved retention, lower leakage, higher local purity, or lower drift.

The transition from broad subspaces to dynamic kernels is central to the later subspace-selective strategy. A uniform search treats all candidate regions as equally important. A kernel-based search instead allocates more attention to regions where several forms of evidence overlap:

$$\text{static separation} + \text{recurrence} + \text{anchor support} + \text{dynamical support}.$$

This is the conceptual basis for the v24_8-style direction of the project. The purpose of later versions is not merely to confirm that broad REAL-NULL separation exists. The purpose is to identify smaller regions where the signal is concentrated and where dynamical behaviour is measurably better than in matched controls.

8.4 Physical and computational interpretation

The physical interpretation should remain modest. The results indicate that under the chosen metrics and matched-control construction, selected regions of low-qubit Hilbert space behave differently from their controls. This suggests non-uniform internal structure, but it does not by itself identify a new physical phase, a protected code space, or a complete quantum error-correcting mechanism.

The computational interpretation is more immediate. If candidate soft spaces can be identified reproducibly, then the search problem becomes more efficient. Instead of treating the entire Hilbert space as equally relevant, the method can prioritise subspaces where evidence is concentrated.

This gives the project a practical direction:

identify broad candidates $\implies$ test dynamically
 $\implies$ zoom into kernels
 $\implies$ predict higher-qubit continuations.

The most important implication is therefore methodological. The 4-to-8 qubit track suggests that lower-qubit evidence can be used to guide higher-qubit search. If this continues to hold in later experiments, the project may develop from retrospective detection into a predictive framework for soft-space scaling.

9 Limitations

Several limitations must be acknowledged. These limitations do not invalidate the results, but they define the level of interpretation that is currently justified.

First, the operational definition of relevance R remains under refinement. In the present framework, R should be understood as a metric family rather than as a single final scalar. Different operational definitions may emphasise different aspects of candidate soft-space behaviour, including fidelity-based retention, probability-density concentration, leakage reduction, local purity, structural drift, or computational cost. This means that any claim about relevance must specify which metric is being used.

Second, the dynamical evidence is weaker than the static evidence. The most reproducible finding in the project is static REAL-NULL separation. Dynamical indicators under Lindblad-type evolution are more subtle and more sensitive to modelling choices. They should therefore be interpreted as suggestive rather than conclusive. At the current stage, the dynamical results motivate targeted kernel search, but they do not by themselves establish robust dynamically protected subspaces.

Third, the analysed systems remain low-qubit systems. The study covers

$$n \in \{4, 5, 6, 7, 8\},$$

corresponding to Hilbert-space dimensions

$$16 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256.$$

This is a meaningful scaling track for a Phase 1 investigation, but it does not establish scalability to large quantum processors. Any extension beyond 8 qubits requires further testing, normalisation of metrics across Hilbert-space dimension, and careful control of computational cost.

Fourth, the results depend on the quality of the matched NULL construction. The NULL model is central because it defines the reference against which REAL structure is judged. If the NULL model is poorly matched, it may either exaggerate or suppress apparent REAL-NULL separation. The conclusions are therefore conditional on the assumption that the NULL controls preserve the relevant baseline properties while removing the structure under investigation.

Fifth, the identification of anchor points and hotspots remains operational. A recurrent index pair or neighbourhood, such as a candidate region around

$$(157, 158),$$

should not be interpreted as meaningful solely because of its numerical label. Its significance depends on recurrence, local neighbourhood coherence, static separation, dynamical support, and compatibility with lower-qubit priors. Without these supporting conditions, an anchor may be a sampling artefact.

Sixth, the Lindblad models used in the dynamical analysis represent selected noise assumptions. Different Hamiltonians, Lindblad operators, time horizons, time steps, or noise-qubit selections may change the observed dynamical behaviour. Therefore, conclusions about retention, leakage, purity, or drift must always be tied to the specific open-system model used in the experiment.

Seventh, the present framework does not yet provide a closed-form predictive law. The proposed soft-space scaling rule is an empirical and operational framework. It is designed to guide future experiments by using lower-qubit evidence as a search prior for higher-qubit systems. It should not yet be interpreted as a universal scaling law.

Finally, the present paper should be read as a Phase 1 evidence report. Its strongest contribution is methodological: it defines a reproducible REAL-NULL comparison framework, documents static separation across the 4-to-8 qubit track, and motivates a more selective dynamical search strategy. The next stage must test whether the candidate structures identified here remain stable under broader noise families, additional batches, higher qubit numbers, and more stringent validation criteria.

10 Future Work

The next stage of the project is a systematic subspace-selective scaling study. The central methodological change is to move away from uniform scanning of all candidate regions and toward targeted analysis of regions where the REAL-NULL contrast is large, recurrent, or dynamically suggestive.

The future-work programme has five main components.

10.1 Hierarchical zoom into candidate regions

The first priority is to develop a hierarchical zoom procedure. Broad candidate regions should be identified by static REAL-NULL separation and recurrence. These broad regions should then be tested dynamically under Lindblad-type evolution. If a broad region shows favourable behaviour, the method should zoom into smaller internal subregions.

This can be written schematically as

$$\mathcal{S}_{n,j} \supset \mathcal{S}_{n,j}^{(1)} \supset \mathcal{S}_{n,j}^{(2)} \supset \mathcal{K}_{n,j},$$

where $\mathcal{S}_{n,j}$ is the original broad candidate region and $\mathcal{K}_{n,j}$ is a smaller candidate dynamic kernel.

The purpose of this procedure is to determine whether the strongest dynamical behaviour is concentrated in smaller sub-subspaces rather than distributed across the entire candidate region.

10.2 Dynamic-kernel identification

A key objective of later versions is to identify dynamic kernels. A dynamic kernel is a subregion that satisfies stronger criteria than a broad soft-space candidate. It should show not only static separation, but also favourable behaviour under dynamical metrics.

The preferred pattern is

$$\Delta R_{n,j} > 0,$$

$$\Delta \Lambda_{n,j} < 0,$$

$$\Delta P_{n,j} > 0,$$

and

$$\Delta D_{n,j} < 0.$$

Thus, a candidate kernel should retain more probability mass, leak less, maintain higher local purity, and drift less than its matched NULL counterpart.

The next versions should therefore test whether broad candidate soft spaces contain smaller kernels where these dynamical advantages are stronger and more stable.

10.3 Extension of the qubit sweep

A future controlled qubit sweep should extend the current 4-to-8 qubit track. A natural next range is

$$n \in \{4, 5, 6, 7, 8, 10, 12\}.$$

This would allow the project to test whether the patterns observed in the low-qubit regime persist as the Hilbert-space dimension grows further:

$$2^4 \rightarrow 2^5 \rightarrow 2^6 \rightarrow 2^7 \rightarrow 2^8 \rightarrow 2^{10} \rightarrow 2^{12}.$$

Equivalently,

$$16 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 1024 \rightarrow 4096.$$

This extension would be important because the transition beyond 8 qubits tests whether the observed structure remains meaningful when the search space becomes substantially larger.

However, comparisons across qubit number require normalised metrics. A raw score at 4 qubits cannot automatically be compared with a raw score at 12 qubits. Future versions should therefore define dimension-normalised versions of static separation, retention, leakage, purity, drift, recurrence, and anchor compatibility.

10.4 Validation across noise models and batches

The dynamical conclusions should be tested across broader noise assumptions. Later versions should compare multiple Lindblad-type noise regimes, including dephasing-style channels, amplitude-damping-style channels, and mixed noise models.

For each candidate region, the relevant question should be whether the favourable direction survives changes in the noise model:

$$\Delta R > 0, \quad \Delta \Lambda < 0, \quad \Delta P > 0, \quad \Delta D < 0.$$

A candidate that performs well under only one very specific noise setting should be treated cautiously. A candidate that remains favourable across multiple noise regimes is substantially stronger.

The same principle applies to batch testing. Candidate anchors, hotspots, and kernels should be required to recur across independent batches or repeated runs. A practical rule for future versions is that a candidate should not be treated as strong unless it shows consistent sign and recurrence across multiple independent evaluations.

10.5 Refinement of the relevance metric

The relevance metric R remains an important open component of the framework. Future work should compare alternative operational definitions, including

- (1) fidelity-based retention;
- (2) probability-density concentration;
- (3) leakage reduction;
- (4) local purity preservation;
- (5) drift reduction;
- (6) computational cost reduction;
- (7) composite relevance scores.

A composite relevance score may take the form

$$R_{n,j}^{\text{comp}} = a_1 R_{n,j} - a_2 \Lambda_{n,j} + a_3 P_{n,j} - a_4 D_{n,j} + a_5 B_{n,j},$$

where $B_{n,j}$ denotes recurrence or batch stability, and the weights $a_1, \dots, a_5$ must be calibrated empirically.

The purpose of refining R is to make the framework more predictive. A well-defined relevance score should help decide which lower-qubit structures should be used as priors for higher-qubit searches.

10.6 Anchor continuation and predictive testing

Future versions should also test anchor continuation more explicitly. If a candidate anchor appears at n qubits, the method should generate a predicted search region at $(n+1)$ qubits before the full higher-dimensional scan is evaluated.

This converts the framework from retrospective analysis into predictive testing:

$$\text{observed anchor at } n \implies \text{predicted region at } (n+1).$$

The prediction can then be tested against the actual higher-qubit results. A successful prediction would mean that the higher-qubit structure appears near the expected anchor neighbourhood, or that it preserves the expected signature, recurrence, or dynamical behaviour.

This is particularly important for 8-qubit and later anchor regions such as neighbourhoods around

$$(157, 158).$$

Such regions should be treated as hypotheses for further testing, not as final conclusions.

10.7 Toward a soft-space atlas

The long-term goal is to build a soft-space atlas: a table-like organisation of candidate structures across qubit number, anchor family, static evidence, dynamical evidence, and recurrence.

Such an atlas could have the form

Qubits	Anchor	Static score	Dynamic score	Recurrence
4	a_4	$[\cdot]$	$[\cdot]$	$[\cdot]$
5	a_5	$[\cdot]$	$[\cdot]$	$[\cdot]$
6	a_6	$[\cdot]$	$[\cdot]$	$[\cdot]$
7	a_7	$[\cdot]$	$[\cdot]$	$[\cdot]$
8	a_8	$[\cdot]$	$[\cdot]$	$[\cdot]$
10	a_{10}	$[\cdot]$	$[\cdot]$	$[\cdot]$
12	a_{12}	$[\cdot]$	$[\cdot]$	$[\cdot]$

This would not be a periodic table in the chemical sense. Rather, it would be an empirical atlas of recurring subspace structures and their behaviour under scaling.

The future direction of the project is therefore clear: use the 4-to-8 qubit evidence trail as the basis for a more predictive, hierarchical, and dynamically selective search strategy. Later versions should determine whether the structures identified here can be refined into stable kernels, extended to higher qubit numbers, and organised into a reproducible soft-space scaling framework.

11 Conclusion

This paper has presented a staged 4-to-8 qubit investigation of structured subspaces in low-qubit Hilbert spaces. The central aim was to determine whether selected regions of Hilbert space behave differently from matched NULL controls, and whether such regions can be identified, followed, and prioritised as the number of qubits increases.

The strongest empirical result is reproducible static REAL-NULL separation. Across the 4-to-8 qubit track, selected candidate regions, signature families, or anchor neighbourhoods show measurable differences from spectrum-matched controls. This static separation provides the primary evidence for candidate soft-space structure.

The dynamical evidence is weaker, but still informative. Under Lindblad-type open-system evolution, selected REAL-favoured regions show directionally suggestive behaviour, including improved retention, reduced leakage, higher local purity, or lower structural drift relative to matched NULL controls. These effects should not be overstated, but they are important because they indicate where more focused dynamic-kernel searches should be applied.

The main methodological conclusion is that static separation and dynamical testing play different roles. Static analysis identifies candidate soft-space regions. Batch recurrence tests whether those regions are stable rather than accidental. Lindblad-based dynamical analysis then acts as a stricter filter, testing whether selected regions retain favourable structure under noise or dissipation.

The 4-to-8 qubit progression also motivates a more predictive interpretation of the project. Lower-qubit structures should not merely be reported retrospectively. They should be used as priors for higher-qubit searches. In this framework, anchor points, recurrent hotspots, and candidate regions such as 8-qubit neighbourhoods around

(157, 158)

are best understood as hypotheses for further testing rather than as final conclusions.

The overall conclusion is therefore balanced. The results do not establish a complete theory of scalable soft spaces, and they do not demonstrate fault-tolerant logical qubits. However, they do show enough structured REAL-NULL separation to justify the next stage of the project. That next stage should use hierarchical subspace selection to search for smaller dynamically favourable kernels inside statistically favoured regions.

In this sense, the present paper should be read as a Phase 1 evidence base. It establishes the methodological foundation, documents the strongest current signal, identifies the weaker

but important dynamical direction, and motivates a transition from broad detection to targeted prediction.

Acknowledgements

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A Appendix A: Run Registry

This appendix is intended to document the exact numerical runs supporting the paper. The purpose of the run registry is reproducibility: each major claim in the results section should be traceable to a script, command, output file, qubit number, noise model, and version identifier.

Table 5: Template for run registry. Placeholder entries should be replaced with exact filenames, commands, and output logs.

Version	Qubits	Script	Output file	Purpose
v24_3	6	[insert script]	[insert output]	[insert purpose]
v24_7	8	[insert script]	[insert output]	[insert purpose]
v24_8	8	[insert script]	[insert output]	[insert purpose]
v24_14	8	[insert script]	[insert output]	[insert purpose]

For each registered run, the following information should ideally be recorded:

- (1) script filename;
- (2) command-line parameters;
- (3) qubit number n ;

- (4) number of seeds or samples;
- (5) number of batches;
- (6) REAL-NULL construction;
- (7) static metrics computed;
- (8) dynamical metrics computed;
- (9) noise model, if applicable;
- (10) output filename;
- (11) summary of the main result.

A full run entry may be written in the following form:

$$\text{Run} = (\text{version}, n, \text{script}, \text{command}, \text{output}, \text{metrics}, \text{result}).$$

The run registry should be updated whenever a result from a versioned experiment is used as evidence in the main text. This prevents the article from depending on undocumented numerical impressions and makes it possible to verify which outputs support each claim.

B Appendix B: Metric Definitions

This appendix collects the formal definitions of the main metrics used in the paper. The purpose is to keep the main text readable while making the numerical framework reproducible. Where the implementation uses a more specialised version of a metric, the exact code-level definition should be recorded in the run registry or in a separate implementation appendix.

B.1 Fidelity

For two pure states $|\psi\rangle$ and $|\phi\rangle$, the fidelity is

$$F(\psi, \phi) = |\langle\psi | \phi\rangle|^2.$$

For density matrices ρ and σ , the general fidelity is

$$F(\rho, \sigma) = \left(\text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right)^2.$$

Fidelity measures similarity between quantum states. A value close to 1 indicates high similarity, while a value close to 0 indicates low similarity. In the present framework, fidelity-based quantities may be used either as static overlap measures or as dynamical retention-like measures under open-system evolution.

A REAL-NULL fidelity contrast may be written as

$$\Delta F_{n,j} = F_{\text{REAL}}(n, j) - F_{\text{NULL}}(n, j).$$

The preferred direction depends on the specific use, but when fidelity represents retention or preservation of a desired structure, the favourable direction is

$$\Delta F_{n,j} > 0.$$

B.2 Retention

For a candidate subspace

$$\mathcal{S}_{n,j} \subset \mathcal{H}_n,$$

let

$$P_{n,j}$$

denote the orthogonal projector onto $\mathcal{S}_{n,j}$. If $\rho_n(t)$ is the evolved density matrix, the retention inside the candidate subspace is

$$R_{n,j}(t) = \text{Tr} (P_{n,j} \rho_n(t)).$$

Retention measures how much probability mass remains inside the selected subspace at time t . The corresponding REAL-NULL contrast is

$$\Delta R_{n,j}(t) = R_{\text{REAL}}(n, j, t) - R_{\text{NULL}}(n, j, t).$$

The preferred direction is

$$\Delta R_{n,j}(t) > 0.$$

For time-averaged retention, one may define

$$\bar{R}_{n,j} = \frac{1}{T} \int_0^T R_{n,j}(t) dt.$$

In numerical experiments this is approximated by

$$\bar{R}_{n,j} \approx \frac{1}{m} \sum_{\ell=1}^m R_{n,j}(t_\ell).$$

B.3 Leakage

Leakage is defined as the complement of retention:

$$\Lambda_{n,j}(t) = 1 - R_{n,j}(t).$$

Equivalently,

$$\Lambda_{n,j}(t) = 1 - \text{Tr} (P_{n,j} \rho_n(t)).$$

Leakage measures how much probability mass has left the selected subspace. The REAL-NULL leakage contrast is

$$\Delta \Lambda_{n,j}(t) = \Lambda_{\text{REAL}}(n, j, t) - \Lambda_{\text{NULL}}(n, j, t).$$

The preferred direction is

$$\Delta \Lambda_{n,j}(t) < 0.$$

Thus, a favourable REAL region leaks less than its matched NULL counterpart.

B.4 Purity

The purity of a density matrix ρ is

$$P(\rho) = \text{Tr}(\rho^2).$$

A pure state has

$$P(\rho) = 1,$$

while a mixed state has lower purity. Under open-system evolution, purity can decrease as the system becomes more mixed, although the exact behaviour depends on the Lindblad operators and the initial state.

For a candidate subspace $\mathcal{S}_{n,j}$, the projected and renormalised density matrix is

$$\rho_{n,j}^{\text{proj}}(t) = \frac{P_{n,j}\rho_n(t)P_{n,j}}{\text{Tr}(P_{n,j}\rho_n(t))},$$

provided that

$$\text{Tr}(P_{n,j}\rho_n(t)) \neq 0.$$

The local purity inside the candidate subspace is then

$$P_{n,j}(t) = \text{Tr}\left[\left(\rho_{n,j}^{\text{proj}}(t)\right)^2\right].$$

The REAL-NULL purity contrast is

$$\Delta P_{n,j}(t) = P_{\text{REAL}}(n, j, t) - P_{\text{NULL}}(n, j, t).$$

The preferred direction is

$$\Delta P_{n,j}(t) > 0.$$

B.5 Structure drift

Structure drift measures how much a structural descriptor changes over time. Let

$$S_{n,j}(t)$$

denote a descriptor of the candidate region at time t . This descriptor may represent a signature vector, ranked family profile, overlap pattern, coherence profile, or another implementation-specific summary of local structure.

A generic drift measure is

$$D_{n,j}(t) = d(S_{n,j}(0), S_{n,j}(t)),$$

where $d(\cdot, \cdot)$ is a chosen distance function. Possible choices include Euclidean distance, cosine distance, total variation distance, or another metric appropriate to the structural descriptor.

The REAL-NULL drift contrast is

$$\Delta D_{n,j}(t) = D_{\text{REAL}}(n, j, t) - D_{\text{NULL}}(n, j, t).$$

The preferred direction is

$$\Delta D_{n,j}(t) < 0.$$

A negative value means that the REAL structure drifts less than the matched NULL structure.

B.6 Static REAL-NULL separation

For a generic static metric M_{static} , the static separation is

$$\Delta M_{\text{static}}(n, j) = M_{\text{static,REAL}}(n, j) - M_{\text{static,NULL}}(n, j).$$

The interpretation of the sign depends on the metric. For scores where larger values indicate stronger structure, the preferred direction is

$$\Delta M_{\text{static}}(n, j) > 0.$$

For scores where smaller values indicate better behaviour, such as error, leakage, or drift-like quantities, the preferred direction is negative.

B.7 Batch recurrence

If a candidate region $\mathcal{S}_{n,j}$ is evaluated across B independent batches, its recurrence score is

$$\text{Recurrence}(\mathcal{S}_{n,j}) = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{\mathcal{S}_{n,j} \text{ appears in batch } b\}.$$

A recurrence score close to 1 indicates that the candidate appears consistently across batches. A score close to 0 indicates that the candidate is not stable.

For a directional metric M , sign stability may also be defined as

$$\text{SignStability}(M) = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\left\{\text{sign}\left(\Delta M_{n,j}^{(b)}\right) = s^*\right\},$$

where s^* is the preferred or dominant sign.

B.8 Composite relevance score

A composite relevance score may combine several static and dynamical indicators. One possible form is

$$R_{n,j}^{\text{comp}} = a_1 \Delta M_{\text{static}}(n, j) + a_2 \overline{\Delta R}_{n,j} - a_3 \overline{\Delta \Lambda}_{n,j} + a_4 \overline{\Delta P}_{n,j} - a_5 \overline{\Delta D}_{n,j} + a_6 B_{n,j}.$$

Here, $B_{n,j}$ denotes batch recurrence or stability, and the coefficients $a_1, \dots, a_6$ determine the relative weighting of the components.

This expression should be treated as a framework rather than a final definition. The weights must be calibrated empirically, and the selected terms should be reported explicitly for each version of the experiment.

C Appendix C: Suggested Figures

This appendix lists the figures that should eventually be included in the paper. The purpose of these figures is to make the evidence trail visually clear: from REAL-NULL construction, through static screening, to Lindblad-based dynamical testing and hierarchical kernel search.

- (1) **Experimental pipeline diagram.** This figure should show the full workflow:

$$\begin{aligned} &\text{REAL} \rightarrow \text{NULL} \rightarrow \text{static screening} \\ &\quad \rightarrow \text{candidate subspaces} \rightarrow \text{Lindblad testing} \\ &\quad \rightarrow \text{dynamic kernels.} \end{aligned}$$

- (2) **Hilbert-space dimension growth from 4 to 8 qubits.** This figure should visualise the dimensional progression

$$16 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256.$$

- (3) **Static REAL-NULL separation by qubit number.** This figure should show the main static separation metric across

$$n \in \{4, 5, 6, 7, 8\}.$$

It should make clear whether the static signal persists, weakens, strengthens, or changes structure as n increases.

- (4) **Ranked family signatures for the strongest runs.** This figure should show the highest-ranking signature families or candidate regions for selected versions. It should help identify whether the same or related structures recur across batches or qubit numbers.

- (5) **Dynamical retention curves.** This figure should compare

$$R_{\text{REAL}}(n, j, t)$$

and

$$R_{\text{NULL}}(n, j, t)$$

over time for selected candidate regions.

- (6) **Leakage comparison between REAL and NULL.** This figure should show whether selected REAL-favoured regions leak less than their matched NULL controls:

$$\Delta\Lambda_{n,j}(t) < 0.$$

- (7) **Local purity under Lindblad evolution.** This figure should show whether selected candidate regions maintain higher local purity under open-system evolution:

$$\Delta P_{n,j}(t) > 0.$$

- (8) **Structure drift comparison.** This figure should compare drift between REAL and NULL regions:

$$\Delta D_{n,j}(t) < 0.$$

- (9) **Recurrence map of dominant subspaces across batches.** This figure should visualise which candidate subspaces, signature families, or anchor neighbourhoods recur across batches. It should distinguish one-off high scores from stable recurrent structures.

- (10) **Anchor-point and hotspot map for 8 qubits.** This figure should show the location of important 8-qubit candidate regions, including neighbourhoods around anchor pairs such as

$$(157, 158).$$

The figure should make clear that such anchors are interpreted as recurrent structural coordinates, not as intrinsically meaningful integer labels.

- (11) **Stepwise continuation diagram from 4 to 8 qubits.** This figure should illustrate the proposed continuation chain:

$$\mathcal{S}_4 \rightarrow \mathcal{S}_5 \rightarrow \mathcal{S}_6 \rightarrow \mathcal{S}_7 \rightarrow \mathcal{S}_8.$$

- (12) **Hierarchical zoom diagram for v24_8-style refinement.** This figure should show how a broad candidate region is refined into smaller subregions and finally into a possible dynamic kernel:

$$\mathcal{S}_{n,j} \supset \mathcal{S}_{n,j}^{(1)} \supset \mathcal{S}_{n,j}^{(2)} \supset \mathcal{K}_{n,j}.$$

These figures are not merely decorative. Each figure should correspond to a specific claim in the main text. In the final article version, every major empirical claim should be supported either by a table, a figure, or a clearly referenced output file in the run registry.